Shadow or law? Measured numbers as projections of an invariant whole, and a test of which of them could be fixed by structure

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Abstract

What does a precise measurement measure? We state and test the shadow principle: what physics measures is a projection—a frame-, epoch-, convention-, history- or model-dependent image—of an invariant whole, and the whole, not the image, is primary. Its established instances are textbook physics (rest mass versus frame-dependent energy and momentum; Aharonov–Bohm holonomy versus the local field; the cross-ratio under central projection); its general form is our thesis. Two consequences follow. First, a projection is many-to-one, so measured shadows identify the whole only up to its fibre: a precisely measured number can be fixed by structure only if it is constant across the admissible completions of what is known. We turn this into a procedure applied before any formula is fitted, with seven dispositions, and apply it to 74 entries compiled in our programme, mostly precision constants. Re-graded blind by three AI model families and adjudicated by us, the procedure files 73 of them (61 in our earlier grading) as not candidates for a new structurally fixed value and 40 (44 in the earlier grading) as outputs of a dynamical history; the one law candidate left is a conditional CP prediction, $\delta_{\mathrm{CP}} \in \{0^\circ, 180^\circ\}$, and by the same rule our own electron–gravity relation, $Gm_e^2/\hbar c = (4/3)\,\alpha^{21} e^{-5\alpha/2}$, is filed as an output of the electroweak history until a mechanism bridges it. A blind reliability test with the same three families failed its registered bar (Krippendorff $\alpha = 0.0955$), mainly because one model family skipped the procedure's then under-specified step of writing a quantity in natural dimensionless form; with that step made mandatory and checkable, a separately preregistered repeat on 60 fresh quantities gave $\alpha = 1$ (identical dispositions on every item; the bar for strong agreement is 0.8). Judged at the knowledge state of their time, all six numbers that a later law fixed—among them a hydrogen line ratio and Kepler's exponent $3/2$—were filed law candidates, and all five historical numerical patterns were refused; the annotators recognised the cases, so hindsight may have helped. Second, what produces the shadows is not identified by them. That the universe is blind mechanism arising by chance is therefore not a result of physics; we argue that the lawful, intelligible order of the shadows is better explained if their source—the whole itself—is reason-like, the Logos of the old tradition, than if it is chance, which already presupposes law, or a brute order, which leaves its own intelligibility unexplained. Since physics advanced whenever two names proved to be one thing, we also state a rule for such translations and offer whole, frame and shadow as a common vocabulary in which the philosophical traditions and physics can describe one structure and check each other.

SUMMARY

Thesis and question. We hold that the invariant whole is primary and measured numbers are its projections (H0). The equations stay unchanged; explanatory primacy turns by 180°. Which precisely measured scalars could be fixed by structure, rather than by a frame, epoch, convention or history? Our apparatus audits a named carrier/readout before choosing or fitting its formula, with the target excluded from permitted conditioning inputs.

Role	Measurement can establish	It does not alone establish
Law-candidate	Test a conditional structural relation	Its physical mechanism
Output of history	Constrain a process and its inputs	A history-independent value
Model choice	Constrain a completion or selector	Selection by structure alone
Not yet a quantity	Test a readout once built	A value of an unmapped expression
Convention-bound	Check a defining convention	A numerical discovery
Epoch-dependent	Infer a time/frame reading	A timeless scalar value
Already governed	Test an established result	A new independent law

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Results and standing.

Result	Finding	Standing
H0: shadow principle	Invariant mass, causal speed, holonomy and cross-ratio are the established instances behind the universal whole-first thesis	Instances established; thesis argued
Apparatus (H1)	Fibre constancy is quantity-identifiability; seven first-stop dispositions	Equivalence established; screen failed in v3, strong agreement in v4
Census (H2)	74 rows; three blind families + adjudication: 73 not law candidates (historical 61): 54 projection-type outputs, 11 governed, 8 not yet quantities; one law candidate, the conditional CP prediction $\delta_{\mathrm{CP}} \in \{0^\circ, 180^\circ\}$; 40 history outputs (historical 44); of the 13 rows not refused historically, 10 were law candidates: all ten were our own rows, and nine moved; $\alpha = 0.65$ on 47 rows (descriptive)	In-sample; adjudicated; yes-branch checked on recognised historical cases (v4)
Reliability v3, v4	v3: $\alpha = 189/1979 = 0.0955$: FAIL; controls 6/6 each. On 43/44 unit-bearing items from outside our catalogue, one family stopped at the units instead of applying the under-specified natural-form step; the other two agreed at 0.927 (exploratory). v4 (separate preregistration; natural-form step mandatory and checkable; 60 fresh items): $\alpha = 1$, all 60 identical in all three families; controls 6/6 each	v3 registered FAIL, cause located; v4 STRONG
Historical arm (v4)	Judged at their knowledge state: all 6 pre-law numbers that a later law fixed filed law candidates (majority); all 5 historical numerical patterns refused; every family recognised all 11 cases	Yes-branch reachable; not blind
Other calibrations	Earlier $\alpha = 0.639, 0.389$; 5.0% matched-decoy story rate; null-conditional likelihood-ratio ceiling ≈ 20 (bar ≈ 50); blind rank $20.5/40$; Koide null panels unchanged with the mass ratios removed from the formula vocabulary	Scoped measurements
Kill record	163 nonempty verdicts in a 180-entry index; archive window April 26–June 29, 2026; not independent no-go proofs	Scoped record
H4: the source	Measurement does not identify the source; a reason-like whole (Logos) explains lawful shadows better than chance, which presupposes law, or brute order, which leaves its intelligibility unexplained	Philosophical argument
Common vocabulary	Whole, frame and shadow under a stated translation rule; six-entry table (§10)	Proposal; philosophical translation

Deciding tests. H0's universal extension needs a forward map and new independent shadows; the textbook instances alone do not prove it. H1's operational-runnability claim failed in v3 and passed in the separately preregistered v4; v4 does not rescue v3, both stand in the record. A pre-fixed static formula, adequate null and confirmed independent prediction can overturn a refusal. Independent reversal of most historical refusals challenges H2. H4 is challenged by unique source-identification from shadows, lawful order from chance without presupposed law, or a brute order that explains its own intelligibility.

Why read on; what is open. The principle supplies the question, the apparatus exposes missing premises, and the source argument states what physics leaves undecided. Under the test our own relation is an output of the electroweak history until a mechanism bridges it; its forward predictions stay registered (one, the neutrino mass sum, is currently under pressure within ΛCDM [1]). A blind positive control for the test's yes-branch (cases the annotators cannot recognise), human reliability, forward confirmation of H0, source discrimination and any engineering apparatus are open. The census is programme-built; the calibrations do not prove the thesis.

CONTENTS

Summary 1

I. Introduction: the whole and its shadows 4

II. The shadow principle 5

III. What a shadow cannot tell 7

IV. The apparatus: dispositions and audit before fitting 9

V. Worked cases: invariance is not numerical selection 13

VI. A programme-built census, not a population estimate 15

VII. Calibrations: reliability and explanatory weight 19
1. Decoys and matched simplicity 22
2. Blind lineup: template degeneracy 22
3. A reader's ablation: Koide without mass-ratio atoms 23

VIII. A scoped route-verdict catalogue 23

IX. Why physics stalls where it does 24

X. The source 25

XI. Discussion: scope, use and what would change the conclusion 29

XII. Conclusion 30

Data and code availability 31

Acknowledgments and responsibility 31

References 31

A. Census: adjudicated dispositions 36

B. The v3 operational card and scientific errata 39

C. Reliability scoring and original-reference matrices 40

D. Story grammar, rates and ablation scope 41

E. Closed routes: scope vocabulary and missing partition 43

F. Claim grades and standing 43

I. INTRODUCTION: THE WHOLE AND ITS SHADOWS

Stand between a lamp and a wall. Your shadow moves when you move, stretches when the lamp is low and shrinks when it is overhead. Someone who had only ever seen shadows could become an excellent scientist of them: measure how their height depends on the angle of the light, write equations that predict it, argue about why it behaves so strangely—right about everything measured, and missing the person of fixed height who stands between the lamp and the wall.

Physics already contains the clearest case. The energy E and momentum $\mathbf{p}$ of a body depend on how the observer moves relative to it; two observers obtain different values and both are correct. One quantity, the rest mass m, defined by $m^2c^4 = E^2 - \mathbf{p}^2c^2$, is the same for all of them. $E = mc^2$ is the special case of a body at rest: the invariant length of the energy–momentum four-vector, seen along one observer's time axis. Energy is, in this precise sense, the time-shadow of mass. The theory that established this is a theory of invariants, and it is named after what varies.

We defend the general form of this observation and call it the shadow principle: what physics measures is a projection of an invariant whole, and the whole is primary. The arithmetic of physics is unchanged by it, to the last decimal place. What changes is which object is taken to be the real one—the meaning turns by 180 degrees while the equations stay where they are. We argue that this is exactly where persistent failures of foundational physics sit: in the repeated attempt to derive projections as if they were laws, and to build the whole out of its shadows.

The principle forces a question: given a precisely measured number, is it something the world fixes, or something a frame, an epoch, a convention or a history fixes? An age normalized by a reference time changes with epoch; a proton/electron mass ratio depends on dynamics a kinematic count does not specify; a coupling can obey a running law without being a timeless number. The question is not whether these quantities obey laws. It is whether a specified scalar readout, given specified auxiliary information, could have a unique invariant value before its numerical expression is chosen or fitted. We compose existing tools into one carrier-specific test; no absolute priority for that assembly is claimed.

This question has substantial prior art. Measurement theory distinguishes numerical statements from meaningful empirical relations by admissible scale transformations [2–6]. Falmagne and Doble's programme of meaningful scientific laws explicitly restricts possible law forms a priori [7, 8]; meaningfulness is not merely dimensional analysis [9] and does not alone impose a unique dataset [10]. Our convention audit belongs to this lineage, not to a newly discovered requirement that physics be invariant.

The completion question is quantity-identifiability. Structural identifiability and observational equivalence were formulated by Bellman and Åström and by Rothenberg [11, 12]. Identifiable combinations can exist when individual model parameters are not identifiable [13]; predictions may be constrained without determining the entire parameter vector [14]. Practical identifiability additionally depends on finite-data uncertainty [15]. Physical equivalence-class and inference formulations already exist [16, 17]. The elementary factorization used below is an exposition of this foundation, not a new theorem.

Classifications and inventories of constants likewise predate our census [18–22]. The distinction between calculable and environmental parameters is explicit in cosmology and particle physics [23–25]. Renormalization, universality and decoupling explain how low-energy predictions may lose sensitivity to some microscopic details while retaining others [26, 27]. Parameter-space compression and sloppiness quantify approximate rather than necessarily exact insensitivity [28, 29]. Neither a small sensitivity nor a dimensionless value proves structural forcing.

Numerical coincidences require an explicit selection model [30, 31]. Their history warns

against treating algebraic attractiveness as physical necessity [32, 33]. Robertson's objection to Wyler is particularly relevant: an arbitrary normalization can masquerade as an invariant prediction [34]. Multiple testing and unrecorded analytical choices cannot be removed by asserting that no continuous parameter was fitted [35, 36]. Target–decoy calibration supplies a precedent for measuring the rarity of a selected score under a declared null [37, 38]; it does not automatically supply posterior truth probabilities.

Our own earlier work concerns the phenomenological electron-gravity relation [1, 39]; a separate bounded-grammar study concerns the Koide ratio [40]. Here they serve as cases on which the audit must impose the same restrictions it imposes elsewhere. A relation's numerical agreement, its explanatory account, its physical matching and its future tests are separate evidential questions. This paper does not turn the earlier numerical results into a mechanism.

What we add is the principle stated with its established instances and its mathematical limit; a seven-disposition test applied before fitting; its application to 74 programme-built rows with reopening conditions; calibrations and a scoped record of our own failures; and the consequence for the question of the source. The procedure is the apparatus of the thesis, not a substitute for it. The thesis is argued; each data result carries its own narrower evidential status. The apparatus must earn its usefulness through its calibrations, not through the number of rows it classifies.

II. THE SHADOW PRINCIPLE

We claim that the invariant whole is primary and measured projections are derived, because the established instances reveal the same reversal of explanatory primacy without changing an equation.

A. Statement

Reality, as far as physics can speak of it, has two layers. The first is an invariant whole: the relations that hold for every observer, in every frame, at every epoch—the layer of law, measure and geometry. The second consists of projections: what a particular measurement, made from a particular frame, at a particular epoch, within a particular convention, extracts from the whole—a frame-, epoch-, convention-, history- or model-dependent image of it. Projections are real and are measured correctly; physics on the projection is right. They are derived, not primary. The error the principle names is taking the shadow for the substance: treating a projection as the thing to be explained, and building the fundamental description upward out of projections.

In each established instance below the equations are standard and stay unchanged; what reverses is the reading of which side is primary. The principle claims that this reversal is not a curiosity of a few cases but the rule. It is a statement about primacy, not a claim that measurement is unreal or that the universe is a computer simulation.

B. Established instances and explicitly marked readings

(a) Invariant mass. As in Sec. I, E and $\mathbf{p}$ are frame components; m is their invariant norm in mass units. Read this way, $E = mc^2$ does not say that mass is a store of energy waiting to be released; it says what an invariant looks like along one observer's time axis. This explanatory reading leaves energy conservation and the mass changes of bound configurations untouched.

(b) The invariant speed. c is a property of the causal structure of spacetime—the null cone—and massless excitations travel at it in local vacuum relativistic kinematics because they have no rest mass. Light reveals the geometry; it does not own the speed. Coordinate speeds in curved spacetime and propagation speeds in media need not equal c; an exact SI numeral is a unit convention, not the geometrical claim.

(c) Holonomy and field. In the Aharonov–Bohm arrangement electrons travel where the magnetic field vanishes, yet the interference pattern shifts with enclosed flux. The observable is the gauge-invariant holonomy of the potential around the loop, a property of the whole configuration that no local field value along the accessible paths contains [41, 42]. Its phase is periodic in flux with period h/e; arbitrary imposed flux is not quantized. Superconducting $h/(2e)$ quantization requires a condensate. Gauge-invariant local fields remain physical. The holonomy of one loop does not fix the local field; the holonomies of all loops fix the connection up to gauge [43]. The Aharonov–Bohm point is that the accessible region's loops carry information that the accessible local field does not.

(d) The cross-ratio. A nondegenerate central projection generally changes distances on a line, while its four-point cross-ratio is unchanged [44]. For distinct collinear coordinates,

$$[a, b; c, d] = \frac{(a - c)(b - d)}{(a - d)(b - c)}. \tag{1}$$

Substitution of $x \mapsto (Ax + B)/(Cx + D)$ cancels all transformation factors where defined. It is the quantity every such shadow shares—the exact mathematical model of the principle, not a newly proposed projection model of the universe.

(e) Coarse-graining [reading]. Thermodynamic state variables are many-to-one images of microstates. Entropy measures the information a coarse measurement discards, relative to a specified ensemble and coarse-graining. This is an information-loss reading, not a claim that every entropy is one observer's ignorance [16, 26, 45].

(f) General relativity [reading]. The field equations can be read with geometry on the explanatory side and locally measured stress–energy as what a local frame extracts. The equations do not decide between the readings; invariant geometrical relations are shared. They couple Ricci curvature and stress, not derive all matter from geometry alone: matter laws, Weyl-curvature data and boundary conditions are additional. Holographic duality likewise equates descriptions; it need not assign ontological priority to either.

C. One structure across physics

Table I sets the established instances beside our readings; the status column is part of the claim. In the electromagnetic row, one loop's holonomy does not determine the connection or every local field, whereas the full set of loop holonomies does determine the connection up to gauge [43]: the row names a global/local contrast, not an identity of those readouts.

D. Whole, frame, shadow

One structure, read three ways. The whole is the invariant. The frame is the reference system and metric through which the whole is read. The shadow is the outcome. The whole seen through a frame appears as a shadow. This reading is an interpretation, not a derivation; it states the principle in the vocabulary of a long tradition, in which the invariant whole was called measure: Heraclitus'

TABLE I. The common structure. Established instances and our readings are deliberately distinct; the table is not proof of the universal thesis.

Domain	Invariant (whole)	Projection (shadow)	Status
Special relativity	Rest mass; interval; invariant speed	Energy, momentum; durations, lengths	Established
Electromagnetism	Global holonomy of a configuration	Local field readouts; potential representatives	Established global/local distinction
Projective geometry	Cross-ratio	Distances in image	Established mathematics
General relativity	Geometry: connection, curvature	Local "force"; measured stress–energy	Reading
Thermodynamics	Microscopic reversible dynamics	Entropy; arrow of time	Reading
Quantum theory	The state	The measured outcome	Reading
Cosmology	Timeless lawful structure	"Beginning"; expansion history; epoch values	Speculative reading
Precision constants	Fibre-constant quantities	History outputs; epoch values; conventions	Tested in Sec. VI

cosmos "with measures of it kindling, and measures going out" (B30), Pythagorean number, and measure as the first good in Plato's *Philebus* [46, 47].

E. Status: H0

Instances (a)–(d) are established physics and mathematics; nothing in them needs this paper. The universal form is our thesis, H0. It makes two kinds of checkable commitment: a mathematical one (Sec. III, what a projection cannot reveal) and an empirical one (Secs. IV–VII): most precisely measured numbers should test as projections. In our constructed census 54 of 74 rows stop as outputs of history, epoch readings, conventions or model choices—projections in the sense of this section; 11 are already governed by established theory, 8 are not yet measurable quantities, and one is a conditional law candidate. This is consistent with the thesis in a programme-built sample, not a prevalence estimate or a proof of universal primacy. A specified forward map and independent shadows are the stronger empirical commitment.

III. WHAT A SHADOW CANNOT TELL

Shadows determine a quantity of the whole exactly when it is constant on the admissible fibre; identifying that quantity need not identify the whole.

Let $\mathcal{W}$ be admissible wholes—completions of current theory compatible with the declared postulates P and everything retained as known (K)—and $s : \mathcal{W} \to \mathcal{S}$ send each whole to its measured shadows. A carrier/readout dictionary defines $q(M) = X(M)$. The target is excluded

from the retained conditioning information. At shadow value k,

$$F_k = \{M \in \mathcal{W} : \ M \text{ admissible under } P,\ s(M) = k\}. \tag{2}$$

A quantity q is determined by the shadows if and only if it is constant on every fibre $s^{-1}(k)$; then

$$q \text{ constant on every fibre of } s \quad \Longleftrightarrow \quad q = \bar{q} \circ s, \tag{3}$$

for a function $\bar{q}$ of the shadows alone. Define $\bar{q}(k)$ using any representative of F_k; constancy makes that choice irrelevant. Conversely equal retained data give equal q under factorization. At one fixed k, constancy on that fibre suffices to identify that target value. This is the elementary factorization behind structural identifiability [11–13]; no new theorem is claimed.

Three things follow. (1) A number varying across the completion fibre is a model choice: measuring it locates us within the admissible family, without selecting beyond it. (2) A number fixed only by the history a whole went through is an output: further digits refine the record, not the reason. (3) Only fibre-constant quantities—or quantities singled out by a principle fixed independently and before the data—can be forced by structure relative to retained information.

Three fibres must be distinguished. Convention orbits change representatives, units or gauges of one physical description; history fibres vary value-setting initial conditions or processes compatible with K; completion fibres vary physically different models reproducing K. Attractors can make a quantity constant on a history fibre. An independent principle S, declared without fitting the target, can restrict a fibre until its target image is a singleton; the resulting prediction remains conditional on P, S and the dictionary. Equations and boundary information still have to determine its numerical value.

Identifying q does not identify M. A fibre need not be connected or smooth, so zero local derivatives need not establish global constancy. With noisy observations, replace equality by a specified error/covariance region and examine the target image interval. A narrow finite-data interval is practical identifiability, not proof of exact structural constancy [14, 15]. Nor does the audit decide which admitted theory class is physically correct.

A. The principle predicts its own limit

From inside—from shadows alone—the whole is identified only up to its fibre. No finite set of projections proves which whole casts them unless the shadows jointly pin a fibre-constant structure. This is not a weakness of the thesis; it is its mathematical content, and it applies to us. Our relation

$$\alpha_G = \frac{4}{3}\alpha^{21} e^{-5\alpha/2} \tag{4}$$

is the case in point: α_G—the datum from which the relation was found—fixes the exponent at 21 within that ansatz (exponent 10 would miss the electron-mass readout by nearly twelve orders of magnitude, $\alpha^{5.5} \approx 10^{-11.75}$); this is a consistency of the fit, not evidence for why 21.

Why 21? We hold that the exponent is structural, not fitted. 21 is the dimension of $\mathfrak{so}(7)$, whose heat kernel at the identity yields the power 21 and the exponent $5/2$ of the relation in one object, and of $\mathfrak{so}(5,2)$, the conformal algebra of five-dimensional spacetime; both are real forms of the same complex algebra B_3 [1]. This is our hypothesis, stated as one: it becomes a result when a derivation fixes the time $\alpha/7$ and the prefactor $4/3$ from the structure, the step the October paper names as open.

This comparison within a frozen readout does not select whole-group ontology. A group-theoretic account as simple as ours exists for 5.0% of matched decoys, and three blind model families did not single our formula out of forty (Sec. VII E; rank 20.5 of 40).

A reading of the whole cannot be proved from one shadow. What can decide is a set of new, independent shadows the whole fixes jointly: forward predictions registered before their data [1]. The present source is not identified. The inverse problem can be sharpened by new independent observables or interventions; merely repeating one readout more precisely need not shrink its fibre.

B. Failure is not confirmation

A determinant gives the inverse problem in miniature: $\mathrm{diag}(a, d/a)$ has determinant d for any nonzero a. The product constraint does not select the matrix or its spectrum. Likewise a failed forward map rejects that map, rather than confirming the ontology it was intended to instantiate.

Let O be failed derivations, H_0 our shadow thesis with a specified candidate whole, and H_R a rival with no currently accessible selector. Their failure Bayes factor is

$$B_{0R}(O) = \frac{P(O \mid H_0)}{P(O \mid H_R)}. \tag{5}$$

If failure is comparably expected under both, this factor is near unity. These probabilities have not been measured; this is a conditional likelihood argument. Positive evidence needs a declared structure, forward calculation and previously unused readout not used to choose the map, with independent confirmation and a rival comparison. No successful test of the programme's particular invariant-whole source is established here. A close fit with a free normalization and a structural story cannot replace that test.

IV. THE APPARATUS: DISPOSITIONS AND AUDIT BEFORE FITTING

A. Seven dispositions

The seven labels classify what is missing from a proposed structure-fixed scalar prediction, not the importance of its measurement.

The four principal roles are law-candidate, output of history, model choice and not yet a quantity. Convention-bound, epoch-dependent and already-governed complete the operational vocabulary. A row receives one disposition by the *first stopping step*; subsidiary reasons and postulate-dependence remain separate fields. Otherwise multiple valid descriptions would make label agreement uninterpretable. In particular, a present temperature may be both expressed in conventional units and epoch-dependent: the ordered rule, not the observer's preference, chooses the terminal label.

LAW-CANDIDATE means an operationally mapped scalar survives the ambiguity audit and has a proposed structural-value relation not already accounted for by established theory. Eligibility is not confirmation. OUTPUT-OF-HISTORY means retained structure and inputs leave value-setting history or dynamics unspecified. Such a quantity can be computed conditional on those coordinates; the refusal withholds an explanation of why those coordinates have their values. Dynamics is not itself disqualifying: an established attractor or universality mechanism can remove history dependence.

MODEL-CHOICE means physically different admissible completions give different target values and no independent selector has been supplied. Measurement constrains which completions describe nature; it is *not* measurement of an arbitrary fiction. A gauge choice, in contrast, changes representatives of the same physical configuration and must not be counted as physical model choice. NOT-YET-A-QUANTITY means the proposed empirical map lacks a carrier or readout, or claims a structure absent from that carrier. Biology has physical carriers; a failed proposed genetic-code quotient does not imply otherwise. Pure mathematical numbers can enter physical laws, but an abstract number alone supplies no independent empirical degree of freedom to be measured.

CONVENTION-BOUND catches a defining identity, unit or normalization presented as a numerical discovery. A dimensional target may first be replaced by a naturally related dimensionless readout, named explicitly rather than formed from an arbitrary same-unit denominator. EPOCH-DEPENDENT catches a specified readout that varies with time, observer or slice. Coordinate corrections used to infer a fixed relational target belong to its readout, not to a change of the target itself. ALREADY-GOVERNED records an established conditional prediction, bound or universality result. A distinct derivation can illuminate it; a second numerical restatement is not a second independent empirical confirmation.

H1 (demarcation). We propose that testing convention, history and completion ambiguities, with the target excluded from permitted conditioning data, is a useful and reproducible screen for unique structure-fixed scalar values. Constancy or an independently declared selector makes a *conditional* prediction possible; the audit itself supplies neither mechanism nor truth. The operational claim is exposed to the revised-card agreement test. A low agreement coefficient challenges runnability; a correctly refused target subsequently explained by a pre-fixed static expression, adequate numerical null and confirmed independent prediction challenges the stated refusal or its admissible class.

H2 (census prevalence). Within this programme-built census and static scope, most rows are provisionally not new structural-value candidates. The historical count motivates this restricted hypothesis; it is not a sampling estimate for physics. Reversal of a majority of historical refusals by independent graders would challenge it. Already-governed rows must not be interpreted as quantities that cannot be fixed by structure.

Relation to H0. The apparatus tests the shadow principle's carrier-specific commitments; its mathematics and experimental failures do not by themselves settle universal primacy.

B. Steps and precedence

The operational sequence is as follows (Fig. 1).

0. **Map first.** Name carrier and readout. An unmapped expression stops as not-yet-a-quantity.

1. **Conventions and epoch.** Separate defining identities from empirical targets. Name a naturally related dimensionless form if available. A conventional identity stops as convention-bound; genuine epoch/frame variation stops as epoch-dependent.

2. **History.** Identify which inputs set the value and whether compatible histories move it. The operational card distinguishes established analytic/perturbative predictions from contingent outcomes and separately solved bound states. History-class value-setting inputs propagate their classification unless they cancel exactly. Free standard-theory input parameters are

classified as history under card v3; this is a prospective operational rule, not proof that their ultimate origin must be historical.

3. **Optional structural plug-in.** A named framework may restrict the permitted expression grammar. The programme's representation-theoretic grammar is one such restriction, neither necessary nor sufficient for laws in general. It is excluded from reliability testing. Its apparent persuasiveness is calibrated in Sec. VII E.

4. **Numerical null.** A closeness claim requires the declared search grammar, tolerance and selected-expression rule to be tested on matched wrong targets. Exact conditional symmetry values have no near-match to test; this does not waive empirical testing or selection accounting.

5. **Completion.** Hold P, K and the dictionary fixed. CONSTANT or independently SELECTED yields a conditional law-candidate; an established selecting theory gives already-governed; variation without selection gives model choice.

6. **Freeze and audit independence.** Declare target, permitted inputs, expression policy, trials, falsifier and deciding data before the comparison. Internal registration and public disclosure dates are distinct; already released data make a comparison retrospective. Shared drivers must be named.

7. **Refusal with reopening.** Preserve the reason and the smallest new evidence or premise that could reopen the target. A changed premise is a new proposal, not erasure of the old failure.

C. A resolution-aware rational-density control

For denominator bound Q, count distinct reduced fractions

$$C_Q(x,\sigma) = \sum_{q=1}^{Q} \sum_{p\in\mathbb{Z}} \mathbf{1}\{\gcd(p,q)=1,\ |p/q - x| \le \sigma\}. \tag{6}$$

A local Farey-density approximation is

$$\mathbb{E}C_Q \simeq \frac{3Q^2}{\pi^2}(2\sigma). \tag{7}$$

It is not a uniform pointwise theorem for short intervals near every rational. Our fresh null translates a window uniformly around the unit circle containing all reduced p/q modulo one, $1 \le q \le 500$, $0 \le p < q$, including $0/1$. There are 76,116 such points. For a fixed half-width σ, the exact count-tail is the Lebesgue measure of centers whose circular interval contains at least the observed number. The exact population mean is $2\sigma \times 76{,}116$, distinct from the Farey approximation. An exact rational endpoint sweep integrates this statistic, without a seed, random draws or Monte Carlo error (Table II). Input measurement widths remain fixed, not newly propagated. This is an arithmetic null, not a physical law probability; a count tail tests density, not nearest-expression closeness. Lost historical sham probabilities are not reproduced by this different population.

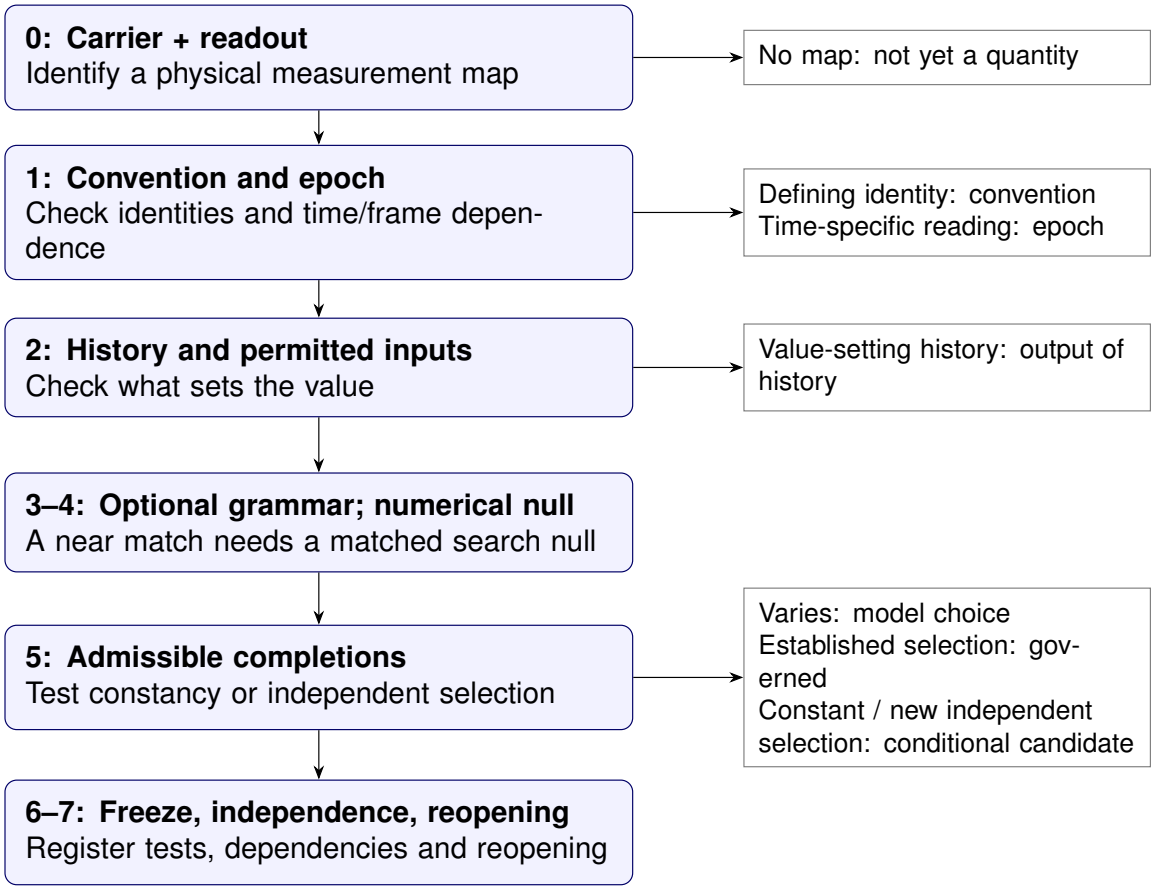


FIG. 1. First-stop audit of a proposed structure-fixed scalar. Numerical screening and the optional framework grammar do not themselves determine a physical value. CONSTANT/SELECTED are conditional on a declared class and dictionary; established explanations take precedence at the final gate.

D. Ten false derivation moves

The following checklist names failure modes, not claims about their frequency outside this programme:

1. Predicting a representation merely because it occurs in a decomposition.
2. Treating a decomposable invariant as one connected realization without constructing it.
3. Reading absolute physical depths from dimensions or Casimirs without a dictionary.
4. Promoting ppm proximity in a scheme-dependent target to an exact law.
5. Mixing amplitude, probability and recorded outcome without a physical instrument.
6. Promoting a tangent or quadratic identity to a finite/all-order law without composition.
7. Copying global charges directly onto local carriers.
8. Treating gauge generators or gauge representatives as direct observables.

9. Selecting a completion because it agrees with an ontology preference.

10. Fitting a determinant and calling its spectrum derived.

A failed forward map is rejected, not converted into confirmation of an interpretation. That evidential guard applies regardless of which programme proposes the map; its likelihood reading is given in Sec. III.

V. WORKED CASES: INVARIANCE IS NOT NUMERICAL SELECTION

Established examples and our own failures distinguish an invariant observable from a selected numerical value.

A. Potential, field and holonomy

For electromagnetism, the potential changes under $A_\mu \mapsto A_\mu + \partial_\mu \chi$, while

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{8}$$

is gauge-invariant and Lorentz-covariant. Its individual components need not be invariant under a change of observer. A potential representative is not a new scalar target; a properly defined field readout is physical. On a multiply connected accessible region, locally vanishing field does not fix all loop phases. The Aharonov–Bohm observable for a charge of magnitude e contains the holonomy [41, 42]

$$W(C) = \exp\left(\frac{ie}{\hbar}\oint_C A \cdot d\ell\right), \qquad \Delta\varphi = \frac{e\Phi}{\hbar} \pmod{2\pi}. \tag{9}$$

Thus h/e is the *period of the phase as a function of flux*, not a quantization of arbitrary imposed flux. Flux quantization at $h/(2e)$ requires the superconducting condensate and its charge and phase-coherence premises; it is not inferred from Eq. (9) alone.

Constancy on a gauge orbit establishes observability, not a numerical prediction. To predict a flux or field value one additionally needs equations, sources, geometry and boundary information. These are already-governed conditional relations. This case also warns against declaring two physical descriptions equivalent merely because a coarse local readout agrees: global holonomy can distinguish them.

B. Two apparatuses, one adjustable driver

An independence error survives entirely different experimental systems. In the plateau-inflation comparison used by the programme,

$$n_s = 1 - \frac{2}{N}, \qquad r = \frac{12}{N^2}, \qquad r = 3(1-n_s)^2. \tag{10}$$

Both proposed values share the free e-fold parameter N. At $N = 57$, $n_s = 55/57$ and the tilt deficit is $2/57$; confusing the last fraction with n_s itself reverses the observable. Counting the two values as independent parameter-free discoveries is wrong. Independent apparatuses can nevertheless test the eliminated relation $r = 3(1-n_s)^2$, subject to the model's approximations and correlated theoretical uncertainties. Dependence in theory does not make all further measurements redundant.

C. Our normalization: a physical completion remains free

Our electron-gravity relation is

$$\alpha_G \equiv \frac{Gm_e^2}{\hbar c} = \frac{4}{3}\alpha^{21}e^{-5\alpha/2}. \tag{11}$$

With CODATA 2022 inputs, it gives $G_{\rm pred} = 6.67431228 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$, $+1.84$ ppm from the recommended G, i.e. 0.08 standard uncertainties of G [48], and 1.16 standard uncertainties above the 2026 NIST redetermination [49]. After the expression search behind it, this match is weak evidence (about 1–2σ in the programme's own look-elsewhere audit [1]). These are numerical context, not a mechanism. For the chosen mathematical readout Z^{-2}, the diagonal heat kernel Z on compact $\mathrm{Spin}(7)$ at $t = \alpha/7$ asymptotically gives $Z^{-2} \propto \alpha^{21}e^{-5\alpha/2}$. There *is* a single mathematical object with that asymptotic form. What is not derived is a physical clock/readout selecting it and the absolute prefactor. In the form given in Ref. [1], the relation equals $(4/3)(7/4\pi)^{21}$ times the inverse square of the $\mathrm{Spin}(7)$ heat-kernel density at $t = \alpha/7$: that readout carries $(4\pi/7)^{21}$ where the relation has $4/3$, a mismatch of about 10^5 ($\frac{3}{4}(4\pi/7)^{21} \approx 1.6 \times 10^5$). For a compact homogeneous readout space with an algebraically fixed invariant metric, a linear clock and the readout Z^{-2}, the normalization-free mismatch is an algebraic number times a nonzero power of $\sqrt{\pi}$, so it is never exactly 1. That is a statement about exact identity: a 22-ppm window cannot tell a rational $4/3$ from such a number, the family is excluded numerically only for the computed cases (each at least 5×10^4 off), and added structure—a further measure, a second scale, a tunable metric or a nonlinear clock—lies outside it. Nor is the exponential completion uniquely selected: five completions with the same first two terms move G between -22 and $+10$ ppm ($+1.84$, $+9.76$, -6.09, $+1.52$, -21.97). All five completions lie within one standard uncertainty of the recommended G ($+0.08$, $+0.43$, -0.27, $+0.07$, -0.98), so G cannot choose among them.

The numerical pieces can be well motivated while their physical assembly remains a choice. Under the first-stop rule the relation already stops at step 2: the electron mass is a free input of the current standard theory, so α_G is an output of the electroweak history until a mechanism bridges it (row c44, Sec. VI)—no exemption for our own formula. Past that gate its normalization would still stop at the completion gate as model choice. Its forward predictions stay registered (one, the neutrino mass sum, is currently under pressure within ΛCDM [1]) and are decided by data. Robertson's analysis of Wyler provides an earlier warning about presenting a free normalization as an invariant prediction [34]. We do not infer that every related mathematical route is impossible. The later predictions were first public on October 6, 2026; internal registrations and retrospective comparisons are kept distinct in the October paper [1].

D. CP: name the symmetry on the observed sector

A real representation or an exchange symmetry alone does not establish the observed leptonic CP phase. The relevant invariant is, for example,

$$J = \mathrm{Im}(U_{e1}U_{\mu 2}U_{e2}^{*}U_{\mu 1}^{*}), \qquad U = U_e^{\dagger}U_\nu. \tag{12}$$

The programme's earlier scan held $U_e = 1$ and interpreted a neutrino-sector reality condition as an unconditional $J = 0$ result. That was a basis artifact: relative charged-lepton structure was omitted from the decisive readout.

The repaired conditional target requires premise X: one common antiunitary CP symmetry on all lepton operators, spurions and vacuum, preserved by light-mode reduction, charged-current

overlap and matching, with an invariant vacuum. Then $J = 0$; where the Dirac phase is physically defined, its CP-conserving values are 0 or 180 degrees. This is a consequence conditional on the complete CP-real structure, not of a representation label or exchange alone [50]. An exchange imposed on the neutrino sector alone, without its relation to the charged-lepton basis, fixes neither J nor $\theta_{23} = 45°$; in that basis exact μ–τ exchange would also force $\theta_{13} = 0$, which is excluded [51]. NuFIT 6.1 (data available in November 2025) still allows CP conservation at 3σ in normal ordering in both tabulated atmospheric-data variants [52, 53]; it excludes $\theta_{13} = 0$. Confirmed CP violation would reject premise X as implemented. The numerical near-match null is vacuous for an exact zero forced by an assumed symmetry, but testing that symmetry is not vacuous.

VI. A PROGRAMME-BUILT CENSUS, NOT A POPULATION ESTIMATE

The census has 74 programme-built rows, mostly precision constants. Re-graded blind by three model families, with disagreements adjudicated by us, 73 of 74 rows are not candidates for a new structurally fixed value; the only law candidate left is a conditional CP prediction, and the test files our own flagship relation as an output of the electroweak history.

A. Historical census

Rows contain named targets and sometimes bundles of related expressions. They are neither independent observations nor a probability sample of all precision measurements. The historical catalogue refused 61 rows in its proposed static-law scope and retained 13. Refusals were 44 history, 6 already-governed, 3 convention-bound, 3 model-choice, 3 without a physical dictionary, and 2 epoch-dependent. The history total is 24 particle/flavour/nuclear/occupancy rows, one materials row, and 19 cosmological rows. All 25 cosmological rows of the historical catalogue were historically refused: 19 history, 3 model, 1 convention and 2 epoch. Appendix A preserves row identities; overlapping composites are not additional tests.

B. Census arm: design

All 74 rows were re-graded with the v3 card. Three model families—OpenAI (gpt-6.1-sol), Google (gemini-3.1-pro) and Anthropic (claude-opus-5-5)—blind to the historical labels and to each other, received only the card and one sealed dossier per row and returned one first-stop disposition per row. A lane is one model family's independent annotation run. The historical census, mapped to v3 labels, was frozen as reference before any lane ran. The 27 rows that appear on the card as worked examples or touch them are reported but excluded from headline agreement because of instructional leakage. The disagreement-adjudication rubric was frozen before the third lane's output was read. This arm is descriptive: it has no pass threshold, is not a registered reliability test, and does not offset the v3 failure (Sec. VII).

C. Agreement

Nominal $\alpha = 4467/6847 = 0.65$ on the 47 rows the card does not use as examples (all 74 rows: 0.68). Per-step agreement is 0.67, 0.67, 0.68, 0.53 for steps 0, 1, 2 and 5. The majority equals the

mapped historical reference on 28 of 47 rows (48 of 74); four rows have no majority. Appendix C gives the non-example matrix and exclusions.

D. Adjudication

All 26 rows where the majority departed from the reference or no majority existed were adjudicated by us—the author's programme—under a rubric frozen before the third lane's output was read; this adjudication is not independent, and the census dossiers and the mapped reference table were written by the builder model (gpt-6.1-sol, the model of lane G). The historical census was at fault on 21; four were ambiguous under the card; one dossier omitted decisive premises; no reference was upheld against an unambiguous, adequately documented reasoned majority. Two printed dispositions keep the historical label against the majority: 3D Ising exponents (card-text conflict, ambiguous) and Feigenbaum's constant (dossier omitted the physical carriers). Both are already governed.

E. Result

Adjudicated census dispositions, with historical exposition counts beside them:

Disposition	Adjudicated	Historical
Output of history	40	44
Already governed	11	6
Epoch-dependent	10	2
Not yet a quantity	8	3
Model choice	2	3
Convention-bound	2	3
Law candidate / historically retained	1	13

The projection-type outputs comprise 54 rows (40 history, 10 epoch, 2 convention and 2 model choice); the other 19 refusals comprise 11 already governed and 8 not yet quantities. Thus 73 of 74 are not candidates for a new structurally fixed value (historical 61). None of the 47 non-example rows is a law candidate.

F. What moved: the test does not spare its authors

Of the 13 historically retained rows, ten map to law candidates on the card (two to output of history, one to model choice); all ten were the programme's own rows, and nine moved. The card's free-input rule—masses, couplings and mixings of the current standard theory are outputs of history until a mechanism derives them from static structure—moves our flagship $\alpha_G = (4/3)\alpha^{21}e^{-5\alpha/2}$, the electron–Planck ratio, the neutrino mass sum Σm_ν ($d = 6$) and the $m_{\beta\beta}$ envelope to output of history. Reopening requires a bridge deriving the input from static structure, then the matched rational-density null. Rows without a physical carrier ($d = 7/8$ dark-sector assignments, ladder exponent, $d = 4$ occupancy, G–α co-variation and method residuals) become not yet a quantity; the large-N ratios R_T, R_J become already governed.

The historical census had exempted its own rows from the gates it applied to the proton–electron mass ratio; the blind lanes did not. The one row that stayed is our conditional CP prediction ($\delta_{\rm CP} \in$

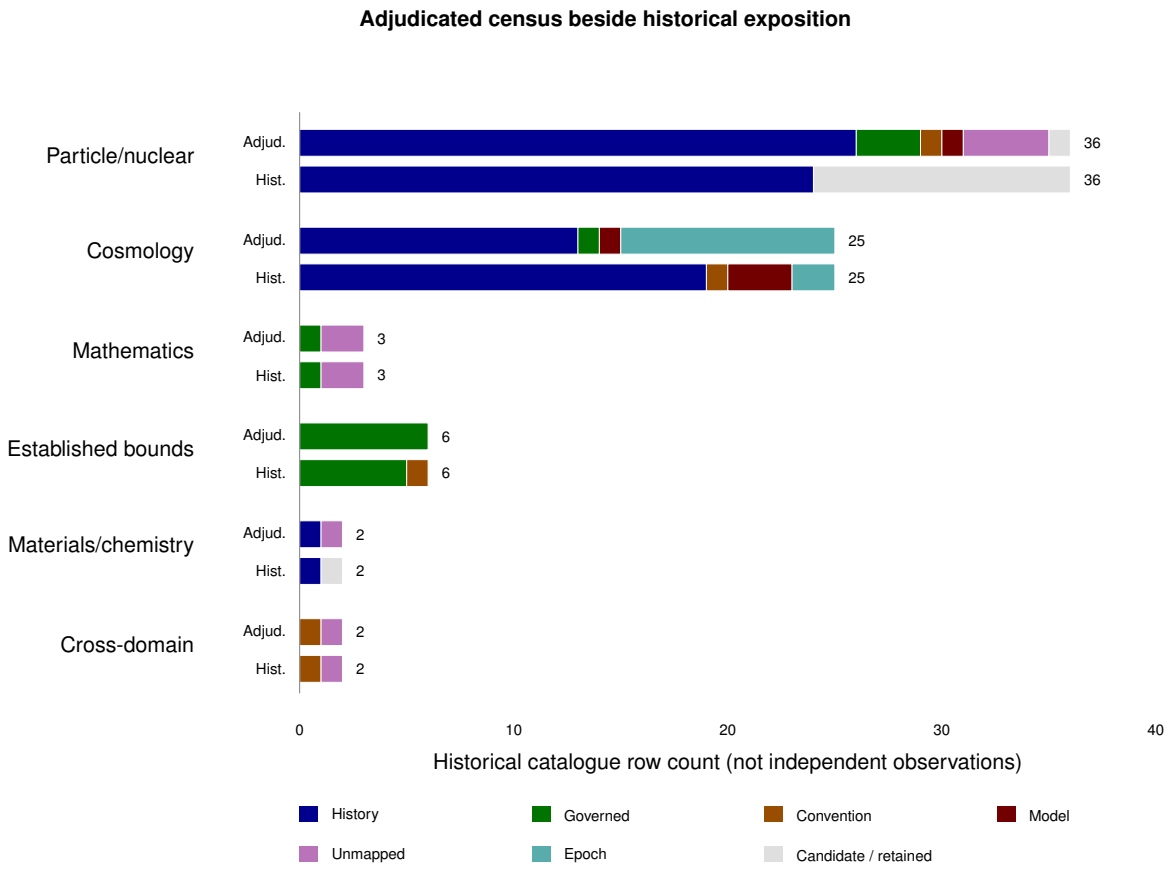


FIG. 2. Adjudicated dispositions by domain, with historical exposition counts beside them. Domain grouping aggregates programme-built row labels, not independent physical degrees of freedom; the same grouping is used for both sets. The historical "retained" class did not certify a law. Row verdicts and leakage markers are in Appendix A.

$\{0°, 180°\}$, $J = 0$ under premise X), which is also the card's own worked example (Sec. V D). Forward predictions of the moved rows stay registered and are decided by data; they are not evidence for a census disposition.

Eight cosmological rows that bundle a present-epoch amplitude stop at step 1: density fractions, the dark-matter/baryon bundle, whose dossier also carried the present-epoch $(\Omega_c + \Omega_\Lambda)/\Omega_b$, $T_{\rm CMB0}$, $\omega_b/\omega_c/\omega_m$, the drag-horizon bundle, σ_8/S_8, photon number density and Ω_K. Their fixed-event components ($z_{\rm eq}$, $r_{\rm drag}$, $\Omega_{\rm DM}/\Omega_b$, η) would stop at history; every cosmological row stays refused. The electron and tau anomalies become already governed by the card's explicit perturbative boundary; the muon anomaly stays history because hadronic inputs set its value at measured precision. $M_{\rm Pl}$ becomes convention-bound; $\rho_\Lambda^{1/4}$ becomes history (free parameter); Bekenstein's 2π and the CMB lensing amplitude A_L become already governed.

G. The yes-branch: historical arm (v4)

Neither the v3 controls (six refusing dispositions) nor any census row is a number known to be fixed by structure, so a 73-of-74 refusal rate alone cannot separate "most precisely measured

numbers are shadows" from "the test rarely says yes." The v4 preregistration (Sec. VII D) added a historical arm for exactly this: eleven measurements presented as they stood at a past state of knowledge, names and dates withheld, judged by the same three families under card v4 *as of* that state. Six were numbers a later law fixed: the 2:1 combining volume of hydrogen and oxygen before Avogadro's hypothesis; the exponent $3/2$ of Kepler's period–distance relation before Newton [54]; the ratio of the two strongest visible hydrogen lines before Bohr [55, 56]; a Landé splitting factor $4/3$ before electron spin [57, 58]; the nuclear closed-shell counts before spin–orbit coupling; and the electron's magnetic-moment anomaly before renormalized quantum electrodynamics. Five were numerical patterns whose non-law nature the knowledge of the time could see: the Titius–Bode growth of planetary gaps [59], the Moon/Sun apparent-diameter ratio, free-fall acceleration in metric units divided by π^2, solar days per year, and the solar-cycle length over Jupiter's period. The majority filed all six law cases law candidate and refused all five patterns (one family filed the closed-shell counts as history under the card's bound-state rule); the preregistered criterion (at least 5 of 6, at most 1 of 5) is met. Every family recognised every case, so knowledge of how the history ended may have helped: the arm shows that the yes-branch is reachable and correctly applied on the classic cases, not that it discriminates blind.

H. The challenge to H2

Independent graders did not reverse the historical refusals: refusals rose from 61 to 73, mainly because the programme's own rows lost their exemption. A majority of independent reversals would have challenged H2; the census arm supplied that challenge and did not reverse them. Because the card refuses free standard-theory inputs by rule and its yes-branch was tested only on recognised historical cases, this was a weak challenge: its outcome is consistent with H2 inside this constructed sample, not a demanding test of it, a prevalence estimate for physics or independent validation of every physical reason.

I. A separate arithmetic null

For the four rational controls, deterministic counting at $Q = 500$ gives Table II. The age observable is $H_0 t_0$, *not* $t_0/t_{\rm P}$; the latter has an entirely different numerical scale. Numbers pass 2 supplies a fresh, exactly evaluated arithmetic null, not a recovered historical sham run. Translate each fixed-width inclusive interval uniformly modulo one around the unit circle, retaining all 76,116 distinct reduced fractions p/q with $1 \le q \le 500$, $0 \le p < q$ (including $0/1$). The tail is the Lebesgue measure of centres whose count is at least the observed count. Rational endpoint sweeps evaluate this measure exactly: no seed, draws, sampling uncertainty or Monte Carlo error. Input widths are held fixed, not assigned an uncertainty by this null. Counts and the exact age null use the unrounded inputs recorded in the number-source register, rather than the shortened table display. The local Farey approximation is not a p-value; the arithmetic tail tests count density under this translated rational grammar, not nearest-expression rarity, history-independence or physical selection. The missing historical sham receipts remain missing.

TABLE II. Exact uniform-circle arithmetic null: observed counts, local Farey approximations, exact null means (rounded here), and count tails. These tails are *not physical law probabilities*.

Target and interval	Count	Farey	Null mean	Tail
$100\theta_* = 1.0411 \pm 0.0003$	45	45.59	45.669600	0.755639
$n_s = 0.9649 \pm 0.0042$	637	638.32	639.374400	0.802831
$H_0 t_0 = 0.95103825 \pm 0.00723112$	1103	1099.00	1100.807716	0.366083
$\Omega_{m0} = 0.315 \pm 0.007$	1073	1063.87	1065.624000	0.096630

VII. CALIBRATIONS: RELIABILITY AND EXPLANATORY WEIGHT

A. The registered reliability results: v3 failed, v4 strong

Two registered results stand side by side. The v3 three-family foreign-target assay **fails** its registered reliability bar: nominal $\alpha = 189/1979 = 0.0955$, below 0.4. The separately preregistered v4 assay, run after the cause was located and the card repaired, is **strong**: $\alpha = 1$ on 60 fresh foreign items, every item filed identically by all three families. v4 does not rescue v3; both stay in the record.

In v3, each family passed all $6/6$ disguised controls; the majority agrees with the pre-output sealed reference on $59/60$ foreign items. Agreement among annotators and agreement with a reference are separate statistics: perfect controls and a largely correct majority do not repair the failed coefficient.

The located cause is mainly one family's non-use of card step 1.3, which permits a standard natural dimensionless form and asks for it when one is known; the pretest had recorded that the admissibility of such forms was not fully operationalized. The cause therefore sits in the instrument as well as in the lane. Our adjudication finds that family applying a FAIL-UNIT template to 43 of the 44 unit-bearing foreign items (45 of 48 including controls). The other two formed admissible natural dimensionless targets. Their pairwise $\alpha = 0.9270$, with per-lane reference agreements $60/60$ and $59/60$, is *exploratory*. The card, pool, dossiers and sealed reference were written by gpt-6.1-sol, the model of lane G; our adjudication ran on the Anthropic family of lane C. The diverging family had no building, reference or adjudication role. Agreement with that reference is therefore not independent of the instrument's authors. Removing the discordant family after seeing its outputs is not the registered three-family test and is never reported as a pass. The operational instrument failed under its actual card-only execution conditions; diagnosis supplies a repair target, not a retrospective change of denominator.

B. Design and scoring

The v3 lanes used OpenAI gpt-6.1-sol (G), Google gemini-3.1-pro (M) and Anthropic claude-opus-5-5 (C), the same versions as the census arm. Each round supplied three different model-family annotators only a procedure card and dossiers for 60 foreign targets plus 6 disguised controls. They returned one first-stop disposition and per-step fields per dossier. Controls were excluded from the reliability and foreign-target validity scores. Nominal Krippendorff agreement uses

$$\alpha = 1 - D_o/D_e, \tag{13}$$

where D_o is coincidence disagreement and D_e is disagreement from the pooled marginals; Appendix C specifies both. This is agreement, not proof that the labels describe physical truth. Preregistered bands were: below 0.4, fail; $[0.4, 0.667)$, inconclusive; at least 0.667, reliable; at least 0.8, strong. They are protocol choices, not general statistical laws. Step 0 in all rounds and

TABLE III. Registered foreign-target agreement and reference hits. Each round has 60 foreign items and three families. v3 final result, cause and exploratory pair are stated before historical diagnostics. v4 used no sealed reference; its validity check is the historical arm (Sec. VI).

Metric	Round 1	Round 2	v3	v4
Final disposition α	0.639	0.389	0.0955	1
Step 1 α	0.757	0.492	0.0955	1
Step 2 α	0.633	0.518	0.0213	1
Step 5 α	0.498	0.469	undef.	1
Step 0 observed agreement	180/180	180/180	180/180	180/180
Controls per lane	5/6	6/6	6/6	6/6
Majority vs sealed reference	44/60	31/60	59/60	no ref.

step 5 in v3 contain a single category (all PASS; all NOT-REACHED on the 60 foreign items), so $D_o = D_e = 0$ and α is undefined; the scorer prints 1 by convention. The only SELECTED-ESTABLISHED item in v3 was a disguised control, excluded from the foreign-item scores. Majority ties count as reference misses. Exact final ratios for rounds 1 and 2 are $13013/20352$ and $8781/22564$.

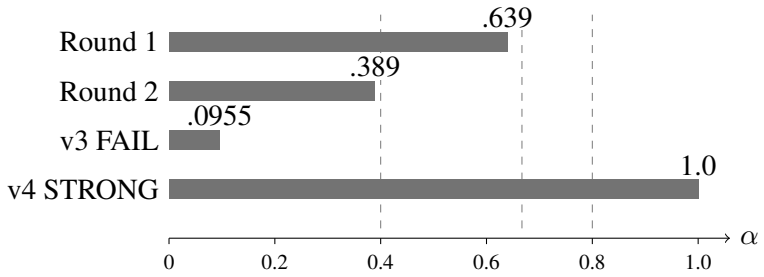


FIG. 3. Final-disposition nominal agreement in the four registered rounds. Lines show the protocol's fail boundary (0.4), reliable threshold (0.667) and strong threshold (0.8). The exploratory two-family v3 coefficient is deliberately excluded. The separate census arm has descriptive $\alpha = 0.65$ on 47 non-example rows, with no pass threshold; it is not plotted on this registered-round scale.

C. What broke and what was repaired

Round 1 was inconclusive. The main boundary disagreement was whether an established but computationally obtained quantity is history or already-governed. In the adjudicated disagreement subset, the author reference was at fault on 14 of 16 items. All lanes missed the same control because it was constructed with the wrong reference role; voiding that control leaves 5/5, which is a correction, not an improved prospective control run.

Round 2 failed the preregistered agreement bar despite perfect controls. "Natural dimensionless form" was read as permission to supply ordinary physics by some lanes and as dossier-only by another; "established" was read as established in physics or explicitly reproduced in the dossier. Input inheritance supplied another recurring ambiguity. Ten reference corrections increased majority agreement from 31/60 to 40/60 (nine additional hits); they do not change the inter-annotator $\alpha = 0.389$. Reference repair after seeing disagreements is useful diagnosis, not sealed prospective validity.

D. What v3 tested, and v4

The v3 card makes natural-ratio permission explicit, gives established-physics precedence, separates perturbative calculation from the contingent/nonperturbative history category, propagates value-setting history inputs, and specifies free standard-theory inputs. The last distinction is an operational convention with physical limits (Sec. XI), not a new no-go theorem. Two comprehension rounds yielded agreement on 5/6 then 4/6 examples; ambiguity was disclosed and the card was not tuned further before the sealed run. The protocol draws 60 foreign targets and 6 controls from a pool of at least 220, excluding the historical catalogue and earlier tested items; a sealed reference precedes lane outputs. Card-only execution forbids lookups. The separate census arm re-grades 74 historical rows and marks worked-example leakage; its outcomes are in Sec. VI ($\alpha = 0.65$ on 47 rows, descriptive; 73/74 refused after adjudication).

The v3 score is final: not pending, bounded provisionally or rescued by control hits. v4 tested the repair prospectively under a separate preregistration (protocol frozen 2026-10-08 before any draw or output, SHA-256 99ff717a…): card v4 is card v3 with the natural-form step made mandatory and checkable—a new field records the ratio, the reference and the physical relation, or the physical reason none exists; "it has units" is not a reason—plus the five fixes the census arm named and a statement of what CONSTANT means when no theory addresses a value. Sixty fresh foreign items were drawn from the v3 pool with fixed quotas per source family (CODATA 10, PDG 12, Planck 3, materials 9, atomic 8, nuclear 8, astronomy 10), so that no single table dominates; the six v3 controls were reused. A one-round wording pretest by fresh GPT and Gemini readers on items outside the pool found a natural form, or a stated defining-identity reason, on every unit-bearing item and required no rewording. The same three families then ran isolated, each reading one sealed file. Result: $\alpha = 1$ (exact) for the final disposition and for steps 1, 2 and 5 (step 0: 180/180 observed agreement, a single category); controls 6/6 each; distribution 41 outputs of history, 10 epoch-dependent, 6 convention-bound, 3 already governed. Each family named a natural form for 32 or 33 of the 43 unit-bearing items and stopped the other 6 under the defining-identity rule (atomic units, the Loschmidt constant); none stopped at units without a stated reason. The v4 card, draw and historical dossiers were prepared by us, an Anthropic model, the family of lane C; the card itself was written inside the programme. v4 had no sealed reference: its validity check for the yes-branch is the historical arm (Sec. VI), whose key is history, not a model. Perfect agreement on a draw dominated by outputs of history and conventions shows that the card can be executed consistently; it does not show that the dispositions are physically correct.

These tests use model families, not human physicists, and correlated training or shared priors can make agreement misleading. Dossiers can omit decisive premises. Neither a repaired author reference nor family diversity constitutes independent physical ground truth. Human annotations, targeted expert disputes and preregistered external samples remain needed for H1.

The census arm agreed at $\alpha = 0.65$ on 47 rows (descriptive, under card v3). The v4 historical arm (Sec. VI) is the first positive check of the yes-branch; its acknowledged leak is model

knowledge of history: every family recognised every case.

E. Calibrating a structural story

A coherent explanatory account is evidence only to the extent that comparably simple accounts are rarer for matched wrong targets under a valid null.

1. *Decoys and matched simplicity*

The programme's calibration drew 100,000 log-uniform targets in $[\alpha_G/10, 10\alpha_G)$ and searched a frozen finite grammar of forms $c\alpha^n e^{-k\alpha}$. The power n came from a group dimension and k from a finite menu of clocks and metrics; the rational-prefactor simplicity was matched to the flagship. The primary tolerance was the CODATA G uncertainty, 22.47 ppm. The grammar was frozen with the flagship already known, so this is not a blind prior over stories.

Of 1,614 numerical-hit decoys, 13 had narrow-layer accounts, 87 medium-layer accounts and 140 wide-layer accounts: empirical rates 0.805%, 5.390% and 8.674%. Selected-expression coverage calculated analytically was respectively 1.007%, 4.99985% and 8.02194%. The medium empirical interval is 4.34–6.61% at the recorded 95% confidence convention. The paper's "5.0%" refers to the medium analytic selected-expression rate, not the percentage of all 100,000 draws. The narrow layer was constructed after the flagship was known; its approximately 1% result cannot supply additional evidential weight. Layer and denominator changes must not be hidden inside a favorable headline.

Let E be existence of an equally simple account under exactly that story policy. If the decoy null is a valid comparable chance model, then

$$\mathrm{LR}(E) = \frac{P(E \mid \text{law model})}{P(E \mid \text{matched null})} \leq \frac{1}{R_{\text{sel}}} \simeq 20. \tag{14}$$

The numerator has not been measured and is at most one. This is a conditional ceiling, not demonstrated odds, $P(\text{law})$, or a universal meaning of a decoy rate. The preregistered weighting rule required $R_{\text{sel}} \leq 2\%$ (ceiling approximately 50) *and* a blind top-two ranking by at least two assessors. Multiplying an ordinary value-search p-value by the story rate would double-count uncontrolled dependence; we do not do so. Target–decoy methods motivate the calibration idea, but proteomic false-discovery assumptions do not transfer automatically to narratives [37, 38].

2. *Blind lineup: template degeneracy*

A second test placed the flagship among 39 matched decoys, all with $n = 21$ and description-length ceiling 28. Three blind families—OpenAI gpt-6.1-sol, Google gemini-3.1-pro and Anthropic claude-opus-5-5—with no tools or project context, each wrote one template account for all 40 formulas: a 22-mode classical oscillator with fitted β and ω; no account; an $\mathrm{SU}(7)$ Weyl-denominator order with tuned k and c. A blind gpt-6.1-sol judge with a frozen 0–10 rubric scored every account within a family identically (5, 0, 6). The flagship's mid-rank is therefore 20.5 of 40 for all three families: all 39 decoys tie it, and no family recognized the formula. Test–retest reproduced $6/6$ scores. The first judge run loaded project context in a follow-up turn and was voided, replaced by a one-shot blinded run, not pooled.

This establishes no target discrimination by those assessors under that protocol, not absence of every possible mechanism. Template degeneracy, model preference and the limited account grammar constrain the test. The preregistered no-weight condition was at least three decoys at or above target (or a selected-account rate at least 20%); the lineup meets it. Hence the story adds no weight *under the preregistered rule*, while Eq. (14) retains the distinct null-conditional ceiling.

3. A reader's ablation: Koide without mass-ratio atoms

The earlier bounded-grammar Koide note studied a numerical attractor, not a physical mass mechanism [40]. A reader reasonably asked whether this arose from mass ratios embedded in the search vocabulary. The registered ablation removed all three mass-ratio atoms (proton/electron, muon/electron and tau/electron). Under the audited cap-16 engine, the main, foreign-target and removed-atom grammars produced identical sham flag vectors at the audited budgets 10, 15 and 16. On the primary 1,024-sham panel, 5/1,024 hit at budget 10 and 19/1,024 at budget 16; an independently seeded budget-16 panel gave 8/1,024. Exact expression-ledger comparisons for the main/foreign grammars agreed 6,144/6,144 at budgets 10 and 16; removal changed 122 expressions without changing the sham flags.

This answers the atom question within those panels and budgets: the reported attractor is not driven by those three atoms there. The inexpensive rational literal $2/3$ remains in the grammar. The ablation does not derive lepton masses, does not extend to unaudited larger budgets, and does not invalidate every Koide claim. Retaining this narrow negative result is more informative than either proclaiming a mechanism or discarding the entire subject.

VIII. A SCOPED ROUTE-VERDICT CATALOGUE

The closed-route record is a map of documented attempts inside one programme, not 163 independent impossibility proofs.

The archive window is April 26–June 29, 2026, not the range of dates of the indexed verdicts. The index dates span April 27–June 25; its dated route rows are June 23–25. The working-note index has 180 entries. Excluding 2 audits and 15 empty verdict cells leaves 163 nonempty route-verdict entries, among 178 actual route documents. Thirteen cells contain only a generic finding label and nine a selector-guardrail label; those 22 generic labels do not establish substantive closure by their wording alone. The arithmetic has been independently recounted, but a full new scientific adjudication of each route has not been done for this paper.

The record uses four scope classes: empirically ruled out, incompatible with a named framework, closed within a declared scope, and still live if a premise changes. They are not interchangeable. A counterexample can refute a universal proposition, but a failure inside one carrier, postulate set or expression grammar does not refute neighboring targets outside it. Per-class counts for this index are not recoverable from the stored assignments. An earlier overlapping family audit is not a substitute; Appendix E preserves that limitation rather than fabricating a four-way partition.

The programme also preregistered a search for a second usable invariant and recorded no qualifying result in the tested grammar. Surviving narrative references do not supply a reproducible run-level sham receipt, so this paper assigns no numeric significance to that null. The historical pattern was that scale-running targets resisted static expressions while discrete structural identities fared better in the tested routes. This is an in-programme observation with selection and scope limitations, not a law separating all dynamical and discrete physics. Running quantities

obey renormalization-group laws; universality can produce precise constants through dynamics. Reopening a route requires a specified new carrier, selector, readout or independent datum, not a new verbal description of the same failure.

IX. WHY PHYSICS STALLS WHERE IT DOES

Many physicists describe the decades since the 1970s as a period without a foundational change comparable to 1900–1930 [60, 61]. This is a diagnosis of foundational progress, not a denial of discoveries or calculational advances. The usual explanations—harder problems, instruments at their limits, the sociology of research—are real. The principle adds a structural one. If a large fraction of the quantities measured most precisely are projections, then the numbers one most wants to explain never resolve however many digits are obtained, because they were never the kind of thing that resolves; and the effort to build the fundamental description upward out of them grows steadily more elaborate, because information discarded in the projection has to be recovered by enumeration. The census (Sec. VI) is consistent with that premise: 54 of its 74 rows stop as projection-type outputs (history, epoch, convention, model choice), 11 are already governed, and none of the 47 non-example rows is a law candidate. Blind re-grading by three model families, with disagreements adjudicated by us, refused 73 as new law candidates (historical 61), our own flagship included. This is a diagnosis of where the effort went, not of who made it: on the projection, the physics is right.

Four examples, ranked by strength. First, the fine-structure constant runs with energy scale: $\alpha^{-1} \simeq 137.036$ at zero momentum transfer and about 128 near the Z mass (the five-flavour modified-minimal-subtraction coupling has $\hat{\alpha}^{(5)}(M_Z)^{-1} = 127.930 \pm 0.008$ in the cited review) [48, 62]. Asking why it equals $1/137$ at every scale asks for a scale-dependent projection to be a static value. The invariant content is the running structure; deriving a boundary value together with its running remains a legitimate structural problem. This example does not prove that no theory could select that boundary condition.

Second, the proton–electron mass ratio is an output of confinement dynamics in a theory whose quark masses are free inputs [62]. It is a history rather than a static target under this paper's operational convention; nonperturbative calculability and future derivations of the inputs are not forbidden. Digits of the output do not by themselves identify the mechanism that fixes all the inputs.

Third, *a relabeling, explicitly not a resolution*: the familiar mismatch of order 10^{120} compares a Planck-cutoff estimate of vacuum energy with the cosmological gravitational response [63]. The size depends on cutoff and comparator; it is not a unique quantum-field-theory prediction. Energy is a projection in the principle's reading, while the cosmological constant may belong to the measure of spacetime. We state that reading as a way to say where not to look. Without a quantitative map predicting the measured gravitational response it solves nothing.

Fourth, the dark sector shows the same split, with a measurable exit. About 85% of the matter in the universe is known only through its gravity [64], and four decades of direct searches have not identified a particle [65]. The evidence has two faces. Rotationally supported galaxies behave like a law: 2693 points in 153 galaxies follow one radial acceleration relation set by the baryons and by one acceleration scale, $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ [66]. Clusters and the early universe behave like an additional collisionless component [64, 67]. Particle and modified-gravity programmes each account for one face naturally and reach the other only with further assumptions. In the principle's reading, "missing matter or modified law" is a split between two projections of one structure, and the test locates the question. The dark-to-baryonic ratio is not a target: the census files its present-

epoch form as epoch-dependent and its fixed-event ratio as an output of history (Sec. VI). The law-shaped number is the coincidence $a_0 \approx cH_0/2\pi$, long noted in the MOND literature [68]: it is a law candidate only if a_0 tracks the expansion rate, $a_0(z) \propto H(z)$, and epoch-dependent if a_0 is constant. A resolved radial acceleration relation at $z \approx 2.2$, where $H/H_0 \approx 3.3$, with shared systematics below 0.02 dex, decides between the two. Our own archival attempt did not reach the required calibration precision, so the relation is untested; we state it as a registered question, not a result.

Likewise, a landscape of order 10^{500} vacua is what the principle expects when one enumerates shadows instead of identifying the whole. The quoted size is a model-dependent flux-vacuum estimate, not a counted population of measured worlds [69]. The diagnosis is our thesis, not a result of the census or a refutation of compactification. A successful map might identify equivalent pre-images, but that equivalence must be demonstrated. A hierarchy problem would change only if an independently derived relation explained a ratio rather than merely selecting a small free parameter [70, 71].

X. THE SOURCE

A. What measurement cannot decide

If measurements reach projections and the whole is identified only up to its fibre (Sec. III), the question of what produces the shadows is not answered by them. This applies to every answer at once. In particular, the statement that the universe is a blind mechanism that arose by chance is not a result of physics. It is a thesis about the whole behind the shadows, exactly as undetermined by those measurements as its alternatives. Physics works under methodological naturalism, a rule of method that has served it extremely well; presenting that rule as a finding about the source takes a conclusion about shadows for a conclusion about the whole—the error the principle names, made at the largest scale. A specified source model can of course have testable consequences; the point concerns alternatives left on the same measurement fibre, not immunity from their consequences.

B. Our thesis

We hold that the source of the shadows is the whole itself, and that it is better understood as reason-like order—the Logos—than as chance or as a brute order, for four reasons.

Lawfulness: The shadows are not noise. In every domain physics has examined they obey compact mathematical relations, the same symmetry principles recur, and their invariants are shared by all observers. This concerns the laws and invariant relations, not identity of all frame components or of all random outcomes.

Chance presupposes law: A probability is defined on a space of possibilities that some lawful structure fixes; chance can select within an order, it cannot account for the order it is defined on.

Intelligibility: The structure is graspable by minds: mathematics developed for its own sake keeps fitting it [72]. Einstein called its comprehensibility the eternal mystery of the world: "the eternal mystery of the world is its comprehensibility" [73].

Primacy: On the principle the invariant whole is prior to every frame. A prior order that every frame shares and that is intelligible is what the oldest traditions named the Logos—the measure and reason of things—and what Spinoza called God or Nature, *Deus sive Natura* (*Ethics*, Part IV, preface) [74]. Einstein's 1929 cable, as reported in the New York Times (25 April 1929, p. 60) via the secondary account, reads in English: "I believe in Spinoza's God who reveals Himself in the orderly harmony of what exists, not in a God who concerns himself with fates and actions of human beings" [75].

This is an inference to the best explanation, not a measurement, and we state it as such. Underdetermination by data is not parity of explanation. Both answers are compatible with the same shadows; they differ in what they leave unexplained. Chance must take as given the lawful space of possibilities on which it is defined; on our thesis that space is not given from outside but belongs to the whole's own intelligible order. The preference is abductive—a judgment of explanatory scope and economy [76, 77]—not a likelihood the shadows supply. What would count against it: a demonstration that shadows identify their source uniquely, an account of lawful order from chance that does not presuppose the order it explains, or an account on which a brute order explains its own intelligibility. These are challenges to the argument H4, not numerical likelihood ratios supplied by the rational or story nulls. Nothing in those calibrations estimates the probability of a Logos.

C. The lineage, and what we add

The distinction between an invariant whole and its appearances is old: Parmenides' changeless Being and the way of opinion; the Logos of Heraclitus; Plato's cave [46, 78, 79]; Plotinus' One and its emanations; Brahman and māyā in Advaita Vedānta; the constant Tao beyond every name that can be spoken [80–82]; Spinoza's one substance and its modes; Kant, who placed the invariant forms on the side of appearance [74, 83]; in physics, Bohm's implicate and explicate order [84], Wheeler's "it from bit" [85], Penrose's Platonic world [86]; structural realism [87, 88]. We read these as a lineage of the insight, not as identical doctrines or independent empirical confirmations.

The many names are themselves an old observation. The Rigveda states it: *ekaṃ sad viprā bahudhā vadanti*—that which is, is one; the sages call it by many names (1.164.46) [89]. Heraclitus' one wise thing "is unwilling and willing to be called by the name of Zeus" (B32) [46]; Cheng Yi and Zhu Xi held that principle is one and its manifestations are many, like the one moon seen in every river and lake [90]. Max Planck drew the same identification across the line between science and religion: in a 1937 lecture he held that "nothing prevents us, and our urge towards a unified view of the world demands, identifying the world order of natural science and the God of religion" [91]. Physics has met the same situation from inside, and its largest steps resolved it: the fall of an apple and the orbit of the Moon became one gravitation; electricity, magnetism and light one field [92]; space and time one spacetime—in Minkowski's words of 1908, space by itself and time by itself are to sink into shadows and only a world in itself is to remain [93]; mass and rest energy one quantity; electromagnetism and the weak interaction one electroweak theory [94]. In each case two descriptions were recognised as one thing when a single description reproduced every measurable consequence of both, and often predicted new ones—electromagnetic waves, the W and Z bosons.

We take that as the rule for the vocabulary we propose. A word of a tradition and a word of physics name the same thing only if (i) they hold the same place in the structure—what is prior and what depends on it; (ii) a statement carried across leads to the same expectations about

observations; and (iii) whatever has no counterpart is marked as a gap, not filled. Under this rule we offer three words—whole, frame and shadow—as a common vocabulary in which the lineage above and physics can state one structure and check each other (Table IV): the physicist's invariant is the tradition's One, a frame-dependent value is its appearance, and the comprehensibility that physics presupposes [72] is what Heraclitus called Logos. The vocabulary is a translation, not a proof; its use is that each side's claim becomes testable in the other's terms. The marked gaps—the observer, purpose and value, for which physics has no agreed account—are where joint work would begin.

TABLE IV. One structure, many names: a vocabulary for the traditions and physics. Entries follow the translation rule in the text; they are examples, not complete lists, and a dictionary, not evidence.

Our word	Traditions (examples)	Physics
whole	Parmenides' Being; Plotinus' One; Brahman; the Tao; Spinoza's one substance	invariant: rest mass, spacetime interval, symmetry group, law
shadow	the shadows in Plato's cave; māyā; "through a glass, darkly" (1 Cor 13:12)	frame-dependent value: energy and momentum; the electric and magnetic parts of one field; a value read at one epoch or scale
frame	the blind men and the elephant (Udāna 6.4) [95]; Kant's forms of intuition [83]	reference frame, units, renormalization scale, gauge, epoch
measure	Pythagorean ratios; "by measure and number and weight" (Wisdom 11:20)	dimensionless ratio, e.g. α
whole in the part	Indra's net [96]; Leibniz's monads as mirrors of the universe (*Monadology* 56)	the hologram: each piece reconstructs the whole scene; related non-local observables: Aharonov–Bohm holonomy, entanglement
source	Logos; Nous; "God or Nature"	the comprehensibility of nature, presupposed by physics and not explained by it

Each grasped the essence; none had an apparatus of this form—a carrier-specific way to say, number by number, what in the measured world is the whole and what is its shadow, with declared fibres, calibrated reliability and reopening conditions. That missing part is what this paper begins to supply (Secs. III–VIII): a test applied before fitting, a census, calibrations and a record of our own failures. This is our claim about the proposed synthesis, not a claim that earlier traditions lacked mathematics, that Bohm lacked a physical theory, or that every ingredient originated here. We supply the beginning of one, and that is exactly why it must stay falsifiable: we finish what they started, we do not found a religion. The apparatus is judged by whether it works—and its registered reliability test has failed once; we print that.

D. Objections and replies

Selection: A universe without lawful order would hold no observers, so we must find ourselves where order permits us. That explains an observer's location within possibilities; it does not explain why there is lawful structure at all. A selection account needs an ensemble generated

by some law and a measure that weighs its members. This question is distinct from whether particular constants are fine-tuned for life; the latter debate also depends on probability and observer-selection assumptions [97]. In eternal inflation the choice of regulator changes relative probabilities: the measure problem is not removed by saying "selection," as the review by Freivogel makes explicit [98]. Selection relocates the order; it does not remove it.

Regularity without government: On a Humean reading, laws summarize what happens and nothing governs behind them [99]. That is a coherent rival position, but it leaves the shared exact invariants of observers' shadows as a brute fact. The principle offers an explanation of that fact. Declining to explain it is not another explanation of it. We prefer the account that explains, while recognizing that this preference is philosophical, not a result extracted from the census.

Necessity: "The order is simply necessary; it needs no source." Then necessity is the name of the source, not its absence: an order that is necessary, intelligible and shared by every frame is what the tradition called the Logos and Spinoza called God or Nature, from whose nature everything follows necessarily (*Ethics*, Part I, propositions 16 and 29) [74]. Chance is then no longer a candidate; the question becomes whether that order is brute or reason-like. We hold the second: a brute order owes us no comprehensibility, whereas an order that carries its own reason explains why it is comprehensible, and the order is in fact comprehensible (Intelligibility). By reason-like we mean such an order: one whose reason lies in itself, so that it is intelligible through and through. Whether it is also purposive, as the Stoics held of their Logos [100], this paper leaves open; Spinoza denied it: Nature "does not work with an end in view" (*Ethics*, Part IV, preface) [74]. The mathematical-universe view [101] gives one version of the necessity answer, the order as a mathematical structure, joined to observer selection among all such structures, which is answered under Selection.

Regress: Hume's Philo asks of any ordering mind behind the world: "Have we not the same reason to trace that ideal world into another ideal world, or new intelligent principle?" (*Dialogues*, Part IV) [102]. Philo's own way out is to stop at the world, "supposing it to contain the principle of its order within itself". Our thesis takes that step: the source is not a further thing behind the order but the order itself, the whole taken as primary, so no regress starts. The step is open to the brute-order view as well; what separates the two is not a further cause but the answer given under Necessity.

Gaps: The source is not placed where calculation currently fails. The argument begins with what physics does calculate: lawful relations among projections. The mathematical limit in Sec. III is that a given projection identifies a source only up to its fibre. More informative measurements can refine that fibre, and an injective measurement map would defeat the stated non-identification premise. For a fixed non-injective map, however, better calculation cannot choose between sources that map to the same output. A gap closes as knowledge grows; the fibre does not close merely because calculation improves.

Method: Physics need not invoke a transcendent source to perform this work. Nothing in Secs. III–VIII uses one: the test, census and calibrations are naturalistic and must be judged as such. This section interprets their setting as philosophy, not as an additional physical input. Keeping those layers apart lets the apparatus fail without disguising its failure as a philosophical success.

Testability: This section is not a measurement, and we do not call it one. It is philosophy with stated defeaters (Sec. X B); neither the rational null nor the story null measures its truth. The claim it opposes—that chance as the ultimate source is a finding of physics—also outruns those measurements when the alternatives remain on the same fibre. A specified chance model with differing observable consequences remains testable; the objection concerns the larger interpretation, not stochastic physics.

XI. DISCUSSION: SCOPE, USE AND WHAT WOULD CHANGE THE CONCLUSION

The audit is useful only if it exposes missing premises and accepts counterexamples to its own decisions.

H1 combines established ideas rather than replacing meaningfulness theory, identifiability analysis or effective field theory. It does not decide that running couplings, environmental quantities or frame components cannot enter laws. Nor can an operational "history" label prove that a parameter will never have a structural explanation. The v3 distinction between perturbative formulas and separate nonperturbative/variational computations may improve card reproducibility, but computation type alone is not a fundamental physical obstruction: a target can be calculable and invariant even when its calculation is hard. Free-input inheritance is likewise relative to present retained information. This boundary must be exposed to experts rather than protected by wording.

The census is programme-built and historically graded inside the same research. Its bundled rows, changing proposal scopes and optional framework grammar preclude population claims. The route index contains uneven verdict strings rather than uniform proof objects. The reliability sample is prospective with respect to its annotators, but references were author-built; disagreements already found serious reference errors. Card v3 failed its registered three-family reliability bar ($\alpha = 0.0955$); the located cause was mainly one family's non-use of the permissive natural-form step, which the pretest had flagged as under-specified, so the cause sat in the instrument as well as in the lane. With that step made mandatory, card v4 reached $\alpha = 1$ on 60 fresh items under a separate preregistration; the family that prepared v4 is one of the three lanes, and agreement among model families is consistency, not truth. The separate census arm agreed at $\alpha = 0.65$ on 47 non-example rows (descriptive) and, after adjudication, refused 73 of 74 rows, our own flagship included; the yes-branch has been checked only on recognised historical cases (Sec. VI). Even agreement under a future protocol would show reproducibility, not correctness. Human physicists and independent domain references remain essential next tests.

The optional group-theoretic grammar is outside the reliability assay. Its story test depends on a matched null and frozen account policy, and the flagship was known at grammar construction. The blind lineup measures the behavior of three model families under one prompt; it is not a survey of human explanatory judgment. Failure to distinguish a target in that lineup bars story weight under the rule, not every future mechanistic account.

> **Use before fitting.** Specify the proposed scalar and physical readout; list postulates and permitted information without the target; vary conventions, histories and completions separately; declare the first-stop reason; for a near-match, freeze grammar, tolerance, selection rule and matched null; name shared drivers; register the deciding observation and reopening condition; keep the interpretation separate from the calculation.

A serious counterexample to a refusal would name the retained information, demonstrate constancy or independent selection on the appropriate fibre, and produce a confirmed prediction not

used to select the map. A future model can make a currently contingent quantity predictable; that is a reopened case under new information, not reason to ignore today's missing input. The low registered v3 agreement rejects the tested card's operational-runnability claim under these execution conditions. Independent reversal of most historical refusals would challenge H2 in this census; the census arm did not reverse them. A forward-tested invariant-whole map would supply specific empirical support for H0 beyond its established instances; failed maps would not. H4 is a philosophical argument, and neither a fitted constant nor an arithmetic null measures the source.

An expert contribution need not endorse the programme: a missing physical premise in one dossier, an incorrectly drawn fibre, a meaningfulness counterexample, a reference correction, or a truly independent prediction can improve the result. Refusal with its scope and reopening condition is a useful outcome; a permanently insulated refusal is not.

A. How the order can be used

If the whole is primary, the order the shadows share is usable without new apparatus, in three ways. It says where a law can be sought and where effort is wasted: the test applied before fitting (Sec. IV B) spares derivations of outputs of history, epoch, convention or model choice. It says which predictions to register before their data, because only independent new shadows narrow a fibre (Sec. III). And it says to steer by patterns rather than by parts—a strategy biology already uses: a brief perturbation of the bioelectric pattern that encodes target anatomy permanently rewrites the planarian regeneration pattern [103], and a 24-hour intervention initiated limb regrowth in adult frogs over the following 18 months [104]. Whether physics has a corresponding lever on what projects its measured values is open: we have no apparatus for that and mark it as a direction, not a result. No engineering prescription for physics follows from these examples.

XII. CONCLUSION

We defend the shadow principle: physics measures projections of an invariant whole, and the whole, not its image, is primary. Established instances leave every equation unchanged while reversing which object bears the explanatory weight. The general form is our thesis H0. Its mathematical limit is exact: shadows determine only what is constant across the admissible fibre, and do not choose a whole simply because a fitted scalar is precise.

The apparatus H1 turns that limit into a question asked before fitting, seven dispositions and scoped reopening conditions. The census, re-graded blind by three model families and adjudicated, refuses 73 of its 74 constructed rows (historical 61), our own flagship included: the test does not spare its authors. This is consistent with H2 inside the constructed sample, not as a population estimate. The three-family v3 calibration failed ($\alpha = 0.0955$) because one family did not apply the then under-specified natural-form step; with that step mandatory, the separately preregistered v4 reached $\alpha = 1$ on 60 fresh items, and in its historical arm the card filed all six pre-law numbers that a later law fixed as law candidates and refused all five historical patterns, on cases the annotators recognised. v4 does not turn v3 into a pass; it shows that the repaired instrument is run the same way by three model families and that its yes-branch is reachable. Our own gravity relation is, under the test, an output of the electroweak history until a mechanism bridges it; its forward predictions remain registered, its narrative supplies no weight under the registered rule, and our route catalogue is not a stack of independent impossibility proofs.

The source H4 is argued, not measured. Chance already presupposes the lawful possibility

space within which it operates; the intelligible order shared by the shadows is better understood, we hold, as the mark of a reason-like whole—the Logos—than as an accident or as a brute order that leaves its own intelligibility unexplained. That argument and the historical lineage do not substitute for the apparatus or for new predictions. A declared whole must fix new independent shadows to earn specific physical support. We keep the thesis, its limit and its failed calibration together: the apparatus serves the principle precisely by refusing to turn an attractive shadow into a law.

DATA AND CODE AVAILABILITY

This paper: doi:10.5281/zenodo.23221015.
Companion repository: `https://github.com/oldrich-research/shadow-or-law`.

The Zenodo record holds this paper, its LaTeX source (sections, bibliography and figure files) and the reproducibility release as one archive with per-file SHA-256 checksums. The repository carries the paper with its source and the same curated release except two files over 10 MB, kept out of its git tree: they are in the record's reproducibility archive, listed with their SHA-256 in the repository's `supplement/README.md`, and a rerun from a checkout uses `supplement/` from that archive. The release of each major revision attaches the record's files as assets. The reproducibility release contains the census CSV and the adjudicated census tables; the exact tested card, foreign and census-arm dossiers, sealed references, lane outputs, frozen adjudication rubrics, adjudication tables and the checksum-pinned scorer; the story grammar, search population, selected-hit data and blinded lineup receipts; the scoped route index; the exact rational calculator with the uniform-circle arithmetic-null protocol; and the standalone figure sources. One command, `python -B supplement/reproduce.py`, recomputes from these files the v3 and round-2 reliability coefficients, the census agreement and dispositions, the arithmetic-null counts and tails, the story rates, the lineup ranks and the route-index counts printed here. Historical missing sham receipts and absent per-route class assignments are disclosed, not replaced by synthetic data.

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AI models were used as tools for research, drafting, annotations and checks. They are not authors or independent human experts. The author directed the research, has no formal physics training, and is responsible for the text, mistakes and scope of its claims. The calibrations deliberately expose failures in both model judgments and the author's references. No acknowledgment implies another person's endorsement.

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Appendix A: Census: adjudicated dispositions

The 74 programme-built rows are proposals, not independent experiments. Three blind lanes G/M/C denote OpenAI (gpt-6.1-sol), Google (gemini-3.1-pro) and Anthropic (claude-opus-5-5); the letters are run labels, not company initials. The historical role is the frozen v3-mapped reference, not the exposition's retained class (13 rows, of which ten map to law candidates, two to output of history and one to model choice). Adjudicated dispositions resolve disagreements under the frozen rubric. H=history, G=already governed, E=epoch, N=not yet a quantity, M=model choice, C=convention, L=conditional law candidate. An asterisk marks instructional leakage: the row remains printed but is excluded from headline agreement. "Changed?" compares adjudicated with mapped historical role.

TABLE V: Census: adjudicated disposition, three lane verdicts and mapped historical role for all 74 rows.

ID	Quantity	Historical	Lanes G/M/C	Adjudicated	Changed?
c01	3D Ising / Wilson-Fisher exponents	G	H/G/H	G	no
c02	Top Yukawa / top mass	H	H/H/H	H	no
c03	$\theta_{\rm QCD}$ / strong-CP angle	H	H/H/H	H	no
c04	Scalar amplitude A_s	H	H/H/H	H	no
c05	Neutrino $d6$ assignment, $\sum m_\nu$	L	H/H/H	H	yes
c06	$\alpha_s(M_Z^2)$	H	H/H/H	H	no
c07*	Density fractions Ω_{m0}, $\Omega_{\Lambda 0}$, $z_{\rm eq}$	H	E/E/E	E	yes
c08	Ω_c/Ω_b and $(\Omega_c+\Omega_\Lambda)/\Omega_b$ today (bundle)	H	E/E/E	E	yes
c09	$m_e/M_{\rm Pl}=\sqrt{\alpha_G}$	L	H/H/H	H	yes
c10*	CMB temperature $T_{\rm CMB0}$	C	E/E/E	E	yes

Continued on next page

Census (continued)

ID	Quantity	Historical	Lanes G/M/C	Adjudicated	Changed?
c11	BBN abundances Y_p, D/H, Li/H	H	H/H/H	H	no
c12	Free neutron lifetime	H	H/H/H	H	no
c13	MSS chaos bound	G	G/G/G	G	no
c14*	Deuteron magnetic moment	H	H/H/H	H	no
c15	ω_b, ω_c, ω_m	H	C/E/H	E	yes
c16	$\Lambda_{\mathrm{QCD}}/\Lambda_5$	H	H/H/C	H	no
c17	CMB lensing amplitude A_L	H	M/H/G	G	yes
c18*	Muon anomalous moment a_μ	H	H/M/H	H	no
c19*	Acoustic scale θ_*, l_A	H	H/H/H	H	no
c20*	Quark mass ratios	H	H/H/H	H	no
c21	Z/A as gravitational-source reading	M	M/M/G	M	no
c22*	BAO sound horizon r_{drag}	H	E/H/E	E	yes
c23	Feigenbaum constants δ, α	G	G/N/N	G	no
c24*	Reionization optical depth τ	H	H/H/H	H	no
c25*	CKM angles / Cabibbo angle	H	H/H/H	H	no
c26	N_{eff}, $N_{\mathrm{gen}} = 3$, ΔN_{eff}	H	H/H/G	H	no
c27*	Deuteron/alpha binding energies	H	H/H/H	H	no
c28*	Electron anomalous moment a_e	H	G/M/G	G	yes
c29	Genetic code $64 \to 20 + \mathrm{STOP}$ orbit quotient	N	N/N/N	N	no
c30*	m_p/m_e	H	H/H/H	H	no
c31	Bekenstein–Hawking coefficient $S = A/4$	G	G/G/N	G	no
c32	Spectral running $dn_s/d\ln k$	H	H/H/H	H	no
c33	Higgs mass / m_H/v / λ_H	H	H/H/H	H	no
c34	SI defining constants / quantum-electrical identities	C	C/C/C	C	no
c35	Fermi constant / EW vev, G_F, v	H	H/H/H	H	no
c36	KSS shear-viscosity ratio	G	G/G/N	G	no
c37*	Universe age t_0, t_0/t_{Pl}	E	E/E/E	E	no
c38	M_{Pl}	H	C/H/C	C	yes
c39	Proton/deuteron/alpha charge radii	H	H/H/H	H	no
c40*	Deuteron electric quadrupole moment	H	H/H/H	H	no
c41*	Hubble constant H_0	E	E/E/E	E	no
c42*	Growth amplitude σ_8, S_8	H	E/H/E	E	yes
c43*	Matter–radiation equality z_{eq}, a_{eq}, k_{eq}	H	H/H/H	H	no
c44*	α_G flagship formula	L	H/H/H	H	yes

Continued on next page

Census (continued)

ID	Quantity	Historical	Lanes G/M/C	Adjudicated	Changed?
c45	$d7/d8$ dark-sector assignments	L	N/N/N	N	yes
c46	Dark-energy equation of state w, w_0, w_a	M	M/H/M	M	no
c47*	Spectral index n_s / tilt	H	H/H/H	H	no
c48	P2-Chem GMTKN55 residuals	L	H/C/N	N	yes
c49	Chaitin's Ω_U	N	N/N/N	N	no
c50	Material T_c/E_F, Homes/Uemura ratios	H	H/N/H	H	no
c51*	Baryon-to-photon ratio η	H	H/H/H	H	no
c52*	Tensor-to-scalar ratio r	H	H/H/H	H	no
c53	α	H	H/H/H	H	no
c54	Photon number density / entropy-per-photon	H	E/E/E	E	yes
c55*	Tau anomalous moment a_τ	H	G/M/G	G	yes
c56*	$\delta_{\mathrm{CP}} = 0^\circ/180^\circ$, $J_{\mathrm{CP}} = 0$	L	L/H/L	L	no
c57	Non-Gaussianity f_{NL}	H	H/H/H	H	no
c58*	PMNS angles / TBM texture	H	H/H/H	H	no
c59	Dark-energy scale $\rho_\Lambda^{1/4}$	M	H/H/H	H	yes
c60*	CMB damping scale k_D	H	H/H/H	H	no
c61	Neutron-proton mass splitting	H	H/H/H	H	no
c62	G-alpha co-variation	L	E/L/N	N	yes
c63	d4 rung occupancy	H	N/N/N	N	yes
c64	$\sin^2\theta_W$, M_W/M_Z	H	H/H/H	H	no
c65	$m_{\beta\beta}$ envelope	L	H/H/H	H	yes
c66	Charged-lepton Koide relation, Q	H	H/H/H	H	no
c67*	Proton/neutron magnetic moments	H	H/H/H	H	no
c68*	Triton/helion magnetic moments	H	H/H/H	H	no
c69	$R_T(5) = 22/105$, $R_J(5) = 16/15$	L	G/G/N	G	yes
c70	Pure mathematical constants	N	N/N/N	N	no
c71	Spatial curvature Ω_K	M	E/E/E	E	yes
c72	CHSH / Tsirelson bound $2\sqrt{2}$	G	G/G/G	G	no
c73	Ladder template exponent	L	N/N/N	N	yes
c74	Bekenstein bound coefficient 2π	C	G/G/G	G	yes

The adjudicated changes and their physical reopening conditions are discussed in Sec. VI, especially subsection F. Row names do not endorse old intervals or a physical carrier that has not been identified.

Appendix B: The v3 operational card and scientific errata

This appendix records the operational instrument in compact form. The exact frozen Markdown card accompanies this paper in the reproducibility release; it is an experimental artifact, not the scientific authority for this paper. The research claim has been narrowed by the dated amendment to structure-fixed scalar values, not all laws. The following summaries preserve the decision rules; outdated worked-example prose is corrected explicitly below rather than repeated as a current result.

Inputs and isolation. Each dossier contains name, definition, units, a value with public provenance and neutral standard context. Use only the card and dossiers, except the explicit permission to recall a standard natural dimensionless form. The protocol's delivery addendum permits reads of the one sealed prompt file only; no lookup, other file, web or memory access is allowed. All items require one verdict. The external assay omits framework-specific step 3.

First stop. Step 0 requires physical carrier and readout; failure gives not-yet-a-quantity. Step 1 first stops defining identities/unit choices as convention-bound, then documented epoch/frame changes as epoch-dependent; otherwise a naturally related distinct reference may give a dimensionless target. Sharing units is insufficient. The reference cannot be the same target under another name or projection.

Step 2 boundary. Established closed-form/perturbative prediction from independent inputs, without contingent initial conditions or target fitting, is STATIC. A contingent outcome or separate variational/nonperturbative solution for a bound state, confined spectrum, broken phase or transition is HISTORY. Value-setting history inputs propagate HISTORY unless they cancel exactly; free masses, couplings and mixing inputs of the current standard theory are HISTORY under the operational card. These input rules take priority over the perturbative rule. Unknown derivation by itself does not establish history. This operational boundary is contestable and is not a mathematical no-go theorem.

Step 4 judgment. REQUIRED means the candidate rests on numerical closeness to a simple expression; VACUOUS means an exact conditional discrete value has no near-match statistic; NOT-APPLICABLE means an earlier stop or no simple-expression claim. Annotators do not calculate the null. The research paper distinguishes count density from nearest-expression rarity and from physical selection; a Farey expectation alone cannot declare an individual coincidence noise.

Step 5 and postulates. Fix postulates, dictionary and admissible completions consistent with retained information. CONSTANT or independently SELECTED-NEW gives a conditional law-candidate; SELECTED-ESTABLISHED gives already-governed; VARIES without selection gives model-choice. Established means established in physics even if not reproduced in the dossier. Established-theory precedence prevents constancy alone being counted as a new prediction. Postulate dependence is an orthogonal field of the justification, not another disposition. Later steps after a stop are NOT-REACHED.

Steps 6–7. Register the value/threshold/falsifier before deciding data; report common drivers and a scoped reopening condition. A failed forward map is rejected, not redescribed as evidence for an interpretation. Exact conditional predictions still need premise and empirical tests.

Recording fields. One block per item: item ID; step0 PASS / FAIL; step1 PASS / FAIL-UNIT / FAIL-EPOCH / NOT-REACHED; step2 STATIC / HISTORY / NOT-REACHED; step4 REQUIRED / VACUOUS / NOT-APPLICABLE; step5 CONSTANT / SELECTED-NEW / SELECTED-ESTABLISHED / VARIES / NOT-REACHED; one of the seven dispositions; confidence high/medium/low; shared driver or none; reopening clause or none stated; one-sentence

justification. A natural ratio must be named. Fields cannot be blank.

Consistency. FAIL at step 0 implies unmapped and later NOT-REACHED (step 4 NOT-APPLICABLE). FAIL-UNIT implies convention; FAIL-EPOCH implies epoch; HISTORY implies history. Step 5 then maps its four reached tokens as above. The ten false derivation moves are in Sec. IV B.

Errata on nonoperative worked examples. The card's historical age-null count 1103 refers to $H_0 t_0$, not age in Planck units. Its "best $2/57$" is the tilt deficit, not n_s. Historical sham probabilities lack their original run receipts; this paper does not certify them. Its older assertion that no single object gives the flagship's form is superseded: a compact diagonal heat kernel reproduces the asymptotic dependence, but not its physical clock/readout or prefactor selection. Claims that a history classification forbids every static formula, that a model-choice measurement establishes nothing about nature, or that a second derivation can add nothing are too broad for the paper's narrowed question. A new selector, independent calculation or additional readout can change the information retained. These errata do not silently change the frozen instrument; effects on the assay must be disclosed by us.

Appendix C: Reliability scoring and original-reference matrices

For m_u available nominal ratings on unit u, let n be the total number of ratings and n_c the pooled category counts. We use

$$D_o = \frac{1}{n} \sum_u \frac{2}{m_u - 1} \sum_{i<j} \mathbf{1}(r_{ui} \neq r_{uj}), \qquad D_e = \frac{n^2 - \sum_c n_c^2}{n(n-1)}. \tag{C1}$$

Three complete ratings per foreign item were available in each historical round. Round 1 gives $D_o = 41/180$, $D_e = 1696/2685$; round 2 gives $D_o = 77/180$, $D_e = 5641/8055$. A single-category field has undefined chance-corrected agreement. The independent recomputation parses raw blocks, excludes controls using their keys, and does not import the original scorer.

Pairwise final-disposition coefficients, in fixed family-pair order, are 0.658423, 0.619159 and 0.641483 in round 1 and 0.138862, 0.436528 and 0.567273 in round 2. Each pair uses its own two-rating marginals. These values are descriptive; no family is treated as ground truth. Original-reference confusion matrices follow. L=law-candidate, H=history, M=model-choice, N=not-yet-a-quantity, C=convention, E=epoch, G=already-governed; a dash column means no majority. All seven reference rows are shown, including zero rows. Such class imbalance constrains validity estimates.

TABLE VI. Round 1: rows are original sealed references; columns are annotator majority. Ties are misses.

Reference	L	H	M	N	C	E	G	–
L	0	0	0	0	0	0	1	0
H	0	24	0	0	1	2	0	0
M	0	0	0	0	0	0	0	0
N	0	0	0	0	0	0	0	0
C	0	1	0	0	8	0	0	1
E	0	0	0	0	0	5	0	0
G	0	10	0	0	0	0	7	0

TABLE VII. Round 2: rows are original sealed references; columns are annotator majority. Ties are misses.

Reference	L	H	M	N	C	E	G	–
L	0	0	0	0	0	0	0	0
H	0	21	1	0	8	1	2	0
M	0	0	0	0	0	0	0	0
N	0	0	0	0	0	0	0	0
C	0	0	0	0	5	0	3	0
E	0	0	0	0	0	2	0	0
G	0	7	4	0	2	0	3	1

After ten reference corrections, round 2 majority accuracy is 40/60 rather than 31/60. The original matrices are not overwritten to conceal that change. The completed v3 sealed-reference matrix is below. It records one majority mismatch rather than overwriting the reference with a post-output correction. Non-use of the under-specified natural-form step 1.3 on 43 of 44 unit-bearing foreign items (45 of 48 including controls) is our located cause (Sec. VII); agreement with an author reference is not independent physical adjudication. The separate descriptive census matrix and instructional-leak exclusions follow. The final pairwise coefficients in family order

TABLE VIII. v3: rows are the pre-output sealed reference and columns the three-family majority, as recorded in the reproducibility release; $59/60$ matches, no majority ties.

Reference	L	H	M	N	C	E	G	–
L	0	0	0	0	0	0	0	0
H	0	52	0	0	1	0	0	0
M	0	0	0	0	0	0	0	0
N	0	0	0	0	0	0	0	0
C	0	0	0	0	3	0	0	0
E	0	0	0	0	0	4	0	0
G	0	0	0	0	0	0	0	0

C/M, C/G, M/G are -0.2106, 0.9270, -0.1746; per-family reference hits (C, M, G) are $60/60$, $20/60$, $59/60$. Only the full three-family $189/1979$ coefficient is the registered primary result. The C/G pair is exploratory, not a substituted assay.

Instructional-leak exclusions (27): c07, c10, c14, c18, c19, c20, c22, c24, c25, c27, c28, c30, c37, c40, c41, c42, c43, c44, c47, c51, c52, c55, c56, c58, c60, c67, c68. They remain in Appendix A. No-majority rows: c15, c17, c48, c62.

Appendix D: Story grammar, rates and ablation scope

The primary story assay matches a finite selected-expression policy rather than counting every available account. For a positive reduced prefactor $c = p/q$, the integer code length is $\ell(m) = 2\lfloor \log_2 m \rfloor + 1$, and $\mathrm{DL}(c) = \ell(p) + \ell(q)$. The simplicity-matched panels use $\mathrm{DL}(c) \leq 8$, the cost of $4/3$. The medium layer contains 44 slopes at $n = 21$ from the declared clocks/metrics. A

TABLE IX. Census arm, 47 non-example rows: rows are the frozen v3-mapped historical reference, columns are the blind three-family majority. Matches: 28/47; four ties count as misses. This is not the adjudicated-disposition table.

Reference	L	H	M	N	C	E	G	–
L	0	3	0	2	0	0	1	2
H	0	18	0	1	1	2	0	2
M	0	1	2	0	0	1	0	0
N	0	0	0	3	0	0	0	0
C	0	0	0	0	1	0	1	0
E	0	0	0	0	0	0	0	0
G	0	1	0	1	0	0	4	0

nearest-hit selector can cover a target range differently from the union of all eligible expression windows; the selected rate is the reported statistic.

The archived search population uses $\alpha = 0.0072973525692838015$; its comparison script records 0.0072973525643314245, and the target baseline is $1.7518093998641384 \times 10^{-45}$. Those are frozen historical assay inputs, not the CODATA-context evaluation in Sec. V C; changing them would be a new assay. The log-uniform local range, primary window and selected-hit denominator define the table below. The medium interval is an exact-binomial (Clopper–Pearson) interval

TABLE X. Primary 22.47-ppm window, simplicity-matched selected-account rates among 1,614 hit decoys. Analytic selected coverage and Monte Carlo frequencies are different estimates, not interchangeable numerators.

Layer	Count	Empirical rate	Analytic rate
Narrow	13/1614	0.805%	1.007%
Medium	87/1614	5.390%	4.99985%
Wide	140/1614	8.674%	8.02194%

conditional on the hit count; it does not include uncertainty in the choice of null or grammar. Blind-lineup rankings use mid-ranks for ties, so forty identical scores give 20.5 rather than a privileged rank for the target. The six replay items test scoring stability, not additional targets; their scores are the recorded test–retest result (Sec. VII E).

For Koide, the audited cap-16 panels cover budgets 10, 15 and 16 and compare match-or-beat errors to the real target at the same budget, with equality included. Main (M1), filtered-atom (F) and re-enumerated removed-atom (R) are distinct constructions; equal flags do not imply identical expression pools. The primary panel is uniform on $[0.6, 0.73]$ with seed 20260520; the independent panel uses seed 20260521. Recovered ledgers check the named expression, value, relative error and flag. The small target-serialization difference does not change any flag when the original threshold is reapplied. No unaudited budget-22 claim is inferred.

TABLE XI. Four scope classes used in the notes. No per-class index counts are recoverable. This is not a numeric partition of the 163 verdict cells.

Class	Meaning and reopening
Empirically ruled out	A stated implementation conflicts with a specified comparison. Reopen with a changed implementation or independent correction, not a relabeling.
Framework incompatible	A construction fails its named mathematical/physical premises. Reopen outside that premise set, with the difference explicit.
Scope closed	No admitted construction succeeds inside the searched family. Reopen by changing the family or supplying the missing dictionary.
Live if premise changes	The record itself identifies a premise-dependent opening. This is not an empirical confirmation.

Appendix E: Closed routes: scope vocabulary and missing partition

The recovered index has 180 entries, of which 2 are audits, 15 have empty route verdicts and 163 have nonempty route strings. The 13 generic finding labels and 9 selector-guardrail labels are included in the last count and not upgraded into proved closures. Repeated derivations, shared premises and different checks on the same target are not independent trials. Only a new route-by-route adjudication could make a complete physical scope partition; that work is not replaced by a guessed table in this paper.

Appendix F: Claim grades and standing

The complete register assigns one card per core claim, operational role, methodological guard and historical census row, each with its grade and standing. Grades: thesis = defended philosophical/ontological claim, not measured; philosophical argument = reasoned source claim, not measurement; established = settled result with cited source; derived = explicit elementary argument; recomputed = independently checked arithmetic; measured = recorded scoped experiment; source-bounded = documented but not newly re-derived here; defined = protocol definition; open = no deciding result; falsified = withdrawn earlier claim. Neither measured nor recomputed upgrades a claim into a physical mechanism. Census row cards correspond to Appendix A; core cards are summarized below.

TABLE XII: Core claim register. The complete 164-claim register with grades accompanies the reproducibility release.

ID	Claim	Grade	Standing
C01	The audit concerns structure-fixed scalar values, not whether a quantity can enter any law.	defined	current
C02	Meaningfulness constrains admissible law forms under scale transformations and centrally precedes this audit.	established	current

Continued on next page

Core claim register (continued)

ID	Claim	Grade	Standing
C03	Fibre constancy is equivalent to elementary factorization through retained information.	derived	current
C04	Identifying a quantity does not identify its model; local constancy need not establish global constancy.	derived	current
C05	Finite-resolution identifiability is not exact structural forcing.	established	current
C06	Convention orbits, compatible histories and physical completion fibres must be distinguished.	defined	current
C07	A selector fixed independently can make a target conditionally identifiable.	derived	conditional
C08	A precise scalar can be computable without identifying its value-setting inputs.	derived	current
C09	The composite seven-disposition protocol is proposed and requires external validation.	open	H1 proposed
C10	The rational interval count uses distinct reduced fractions and a local Farey approximation.	defined	current
C11	A count-density tail is not a nearest-expression tail or a probability of a physical law.	derived	current
C12	Field strength is gauge-invariant and Lorentz-covariant, not componentwise frame-invariant.	established	current
C13	Aharonov–Bohm holonomy can distinguish configurations with the same locally vanishing accessible field.	established	current
C14	h/e is the AB phase period, not quantization of arbitrary imposed flux.	established	current
C15	$h/(2e)$ flux quantization requires condensate and boundary premises.	established	conditional
C16	Gauge-orbit constancy establishes an observable but does not select its numerical value.	derived	current
C17	Inflationary n_s and r share N; eliminating N leaves an independently testable conditional relation.	derived	conditional
C18	The best n_s fraction is 55/57; 2/57 is its deficit.	recomputed	current
C19	The flagship expression gives the printed G and its residuals with CODATA 2022 inputs: $+1.84$ ppm (0.08σ) from the recommended value and 1.16σ above the 2026 NIST redetermination.	recomputed	phenomenological
C20	One compact diagonal heat kernel reproduces the flagship dependence asymptotically, not its physical readout or prefactor selection; the prefactor mismatch is about 10^5 and never exactly 1 for algebraically fixed normalizations.	source-bounded	conditional/open mechanism
C21	Five completions with the same first two terms move G between approximately -22 and $+10$ ppm ($+1.84$, $+9.76$, -6.09, $+1.52$, -21.97), all within one standard uncertainty of the recommended G ($+0.08$, $+0.43$, -0.27, $+0.07$, -0.98); normalization remains model choice.	source-bounded	current scope

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Core claim register (continued)

ID	Claim	Grade	Standing
C22	The full electron–Planck ratio inherits free inputs absent a mechanism under v3 rules.	source-bounded	conditional scope
C23	The earlier unconditional zero-J scan was a basis artifact with U_e fixed to identity.	derived	corrected; old unconditional claim withdrawn
C24	$J = 0$ follows conditionally from common CP-real premise X across all lepton data and matching stages.	derived	conditional
C25	Neutrino-sector exchange alone, without a charged-lepton basis, fixes neither J nor θ_{23}; canonical exchange in that basis also predicts $\theta_{13} = 0$, excluded by data.	derived	corrected; old unconditional claim withdrawn
C26	Later predictions were first public October 6, 2026; internal registration is distinct.	source-bounded	current
C27	Historical census totals are 74 rows, 61 refusals and 13 retained rows.	recomputed	historical in-sample
C28	Historical refusals comprise 44 history, 6 governed, 3 convention, 3 model, 3 unmapped and 2 epoch rows.	recomputed	historical in-sample
C29	The history total includes 24 particle rows, one materials row and 19 cosmological rows.	recomputed	historical in-sample
C30	All 25 explicitly enumerated cosmological historical catalogue rows were historically refused, not 25 independent observables.	recomputed	historical in-sample
C31	The four counts and exact uniform-circle arithmetic tails are reproduced; historical sham protocols remain missing.	recomputed	exact arithmetic null, not law probability
C32	The historical age count 1103 refers to $H_0 t_0$, not t_0/t_{P}.	recomputed	corrected
C33	Census arm: $\alpha = .6524$ (47 non-leak rows; all 74: .6751); majority matches historical role on 28/47. We adjudicated 26 disagreements (21 historical faults, 4 ambiguities, 1 dossier defect). Refused: 73/74; projection-type 54, governed 11, unmapped 8; history 40.	measured	descriptive
C34	Historical final nominal agreement is 0.639 and 0.389 in two three-family tests.	recomputed	inconclusive/fail
C35	Step 0 in all rounds and step 5 in v3 have one category (all PASS; all NOT-REACHED on 60 foreign items): $D_o = D_e = 0$, α undefined; scorer prints 1 by convention.	recomputed	corrected
C36	The first round has 5/6 controls per lane; second round 6/6.	recomputed	historical
C37	Round-1 reference was faulted on 14 of 16 adjudicated disagreement items.	source-bounded	historical adjudication
C38	Round-2 reference correction raises majority hits from 31/60 to 40/60 but not inter-annotator agreement.	recomputed	historical adjudication

Continued on next page

Core claim register (continued)

ID	Claim	Grade	Standing
C39	Registered v3 $\alpha = 189/1979 = .0955$: FAIL; controls 6/6 each, majority 59/60. Natural-form non-use on 43/44 unit-bearing foreign items (45/48 including controls); pretest under-specification and author-family confound; pair $\alpha = .927$ exploratory only.	source-bounded	registered FAIL; pair exploratory
C40	Selected medium accounts occur for 87 of 1614 hit decoys, with analytic selected rate 4.99985 percent.	measured	matched-null conditional
C41	Narrow and wide simplicity-matched rates differ and the narrow grammar is post-target.	measured	limited calibration
C42	An account gives a likelihood-ratio ceiling near 20 only under a valid comparable null and fixed story policy.	derived	conditional ceiling
C43	Story and value p-values must not be multiplied as independent evidence.	derived	current
C44	All 39 decoys tie the flagship (rank 20.5/40); scores identical within each family; test–retest 6/6.	measured	no target discrimination
C45	The story has no weight under the registered rule; this does not exclude future mechanisms.	source-bounded	current rule
C46	Koide mass-atom removal leaves audited sham flags unchanged at budgets 10, 15 and 16.	measured	cap-16 only
C47	Koide main and filtered ledgers agree 6144/6144; removed-atom re-enumeration changes 122 expressions but no flags.	measured	cap-16 only
C48	The atom ablation does not derive lepton masses or establish unaudited larger-budget results.	derived	current
C49	The index has 163 nonempty route-verdict strings after removing 2 audits and 15 empty cells from 180 entries.	recomputed	not independent proofs
C50	Twenty-two generic verdict strings do not substantiate closure by wording alone.	recomputed	scope limitation
C51	Per-class route counts and historical second-invariant numerical significance are unavailable.	source-bounded	not recoverable here
C52	The running-versus-discrete pattern is an in-programme observation, not a universal law.	source-bounded	historical scope
C53	Cross-ratio and invariant mass are established instances of H0; their universal extension is the defended thesis.	established	instances established; extension thesis
C54	A determinant does not select a matrix or its spectrum.	derived	current
C55	Many-to-one coarse-graining is established; treating the invariant whole as primary is the defended H0 reading.	established	maps established; reading ours
C56	H0 states the invariant whole is primary; specific physical support beyond its instances needs forward maps and unused shadows.	thesis	defended; no universal empirical proof
C57	Comparable expected failures under rival hypotheses give no favorable Bayes factor; probabilities are not measured here.	derived	conditional
C58	Vacuum mismatch is a Planck-cutoff heuristic; interpreting it or the landscape as shadows is a thesis reading, not a solution.	philosophical argument	reading, not resolution

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Core claim register (continued)

ID	Claim	Grade	Standing
C59	Historical whole/appearance lineage is our argued reading, not equivalence of doctrines or empirical confirmation.	source-bounded	historical context
C60	Computation type and current free inputs are operational boundaries, not impossibility proofs against future structural explanations.	derived	current
C61	H2 is only a claim about this constructed census under its declared scope.	open	provisional in-sample
C62	Identifiers and the reproducibility release are listed under Data and code availability.	defined	current
C63	H4: the source of the shadows, the whole itself, is better understood as reason-like order (Logos) than as chance or brute order; argued, not measured.	philosophical argument	defended; not measurement
C64	Whole, frame and shadow are one structure read three ways; an interpretation, not a derivation.	thesis	current
C65	Acting on what projects the world is an open direction; no intervention apparatus or result is claimed.	open	not engineering result
C66	Our flagship is output of the electroweak history under the free-input rule until a mechanism bridges it (c44); forward predictions stay registered (the neutrino mass sum is currently under pressure within ΛCDM).	source-bounded	scoped refusal
C67	Neither v3 nor the census arm contains a positive control; the v4 historical arm files 6/6 pre-law law numbers law candidate and refuses 5/5 patterns, on cases the annotators recognised.	measured	measured (v4)
C68	The separately preregistered v4 assay gives $\alpha = 1$ on 60 fresh items, all filed identically by three families; controls 6/6 each. v4 does not rescue v3.	measured	registered STRONG
C69	The v4 historical criterion is met: $S = 6/6$, $F = 0/5$ (bars $S \geq 5$, $F \leq 1$); every family recognised all 11 cases, so hindsight may have helped.	measured	historical criterion met
C70	The two-faced dark sector is read as projections of one structure (thesis); $a_0(z) \propto H(z)$ versus constant a_0 is a registered question, untested after archival calibration failure.	thesis	registered question; untested
C71	Whole, frame and shadow translate traditions and physics under a stated rule: structural place, observational expectations and marked gaps; six-entry table, not proof or evidence.	philosophical argument	proposal; translation
C72	Use the order before fitting and to register independent predictions; cited bioelectric/intervention biology motivates an open physics direction, not a demonstrated physical lever or engineering prescription.	source-bounded	cited biology; physics direction open
C73	Gravitation, electromagnetism and light, spacetime, mass and rest energy and electroweak theory are cited historical unifications in which one description reproduced the consequences of two; Planck (1937) identified the world order of science with the God of religion. They motivate the proposed vocabulary, not a new derivation.	established	cited history